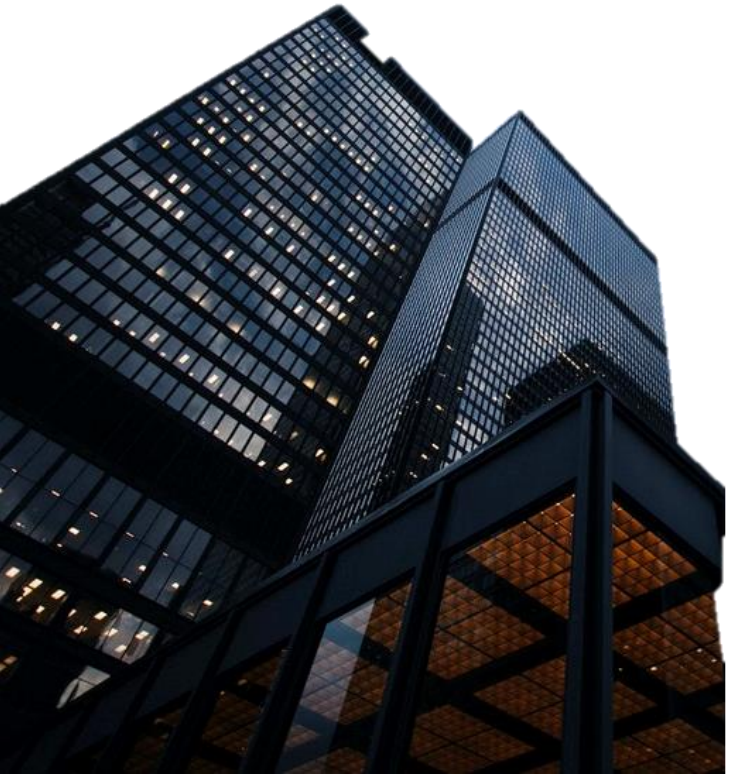
CE 208
Quantity Surveying

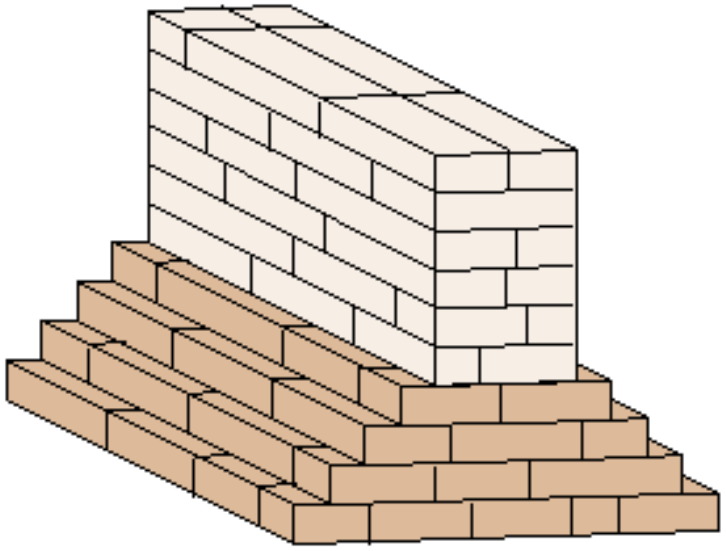
Estimation of a Residential Masonry Building



DEPARTMENT OF CIVIL ENGINEERING
BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Brick Wall Foundation

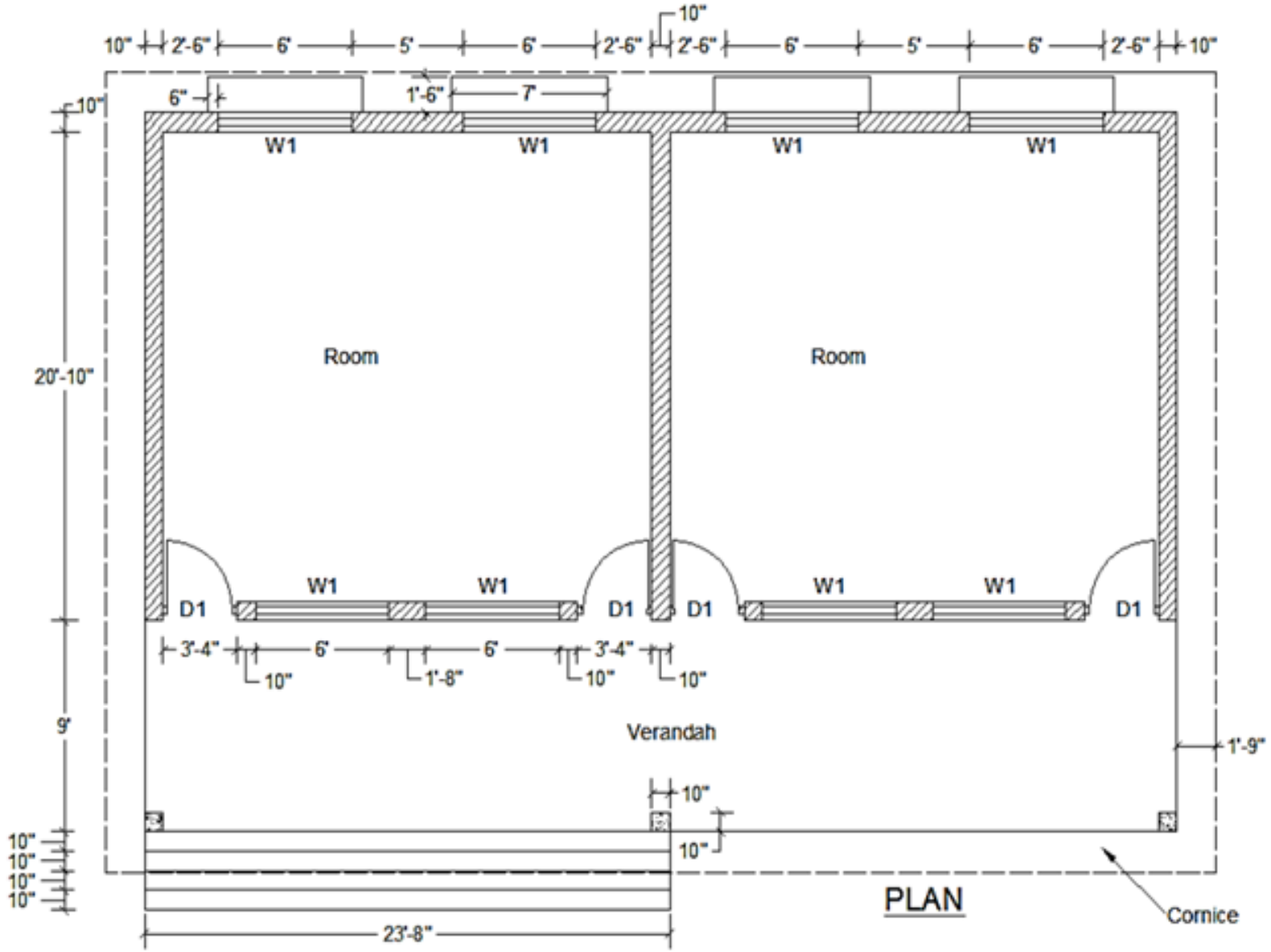
- Wall footing runs along the direction of the wall.
- The size of the footing and the thickness of the foundation wall are specified on the basis of the type of soil at the site.
- The wall footing can be constructed from stone, brick, plain concrete, or reinforced concrete.
- In the case of brick walls, the footing consists of several courses of bricks, the lowest course being usually twice the breadth of the wall above.
- The increased base width of the wall footing is obtained through the provision of 2.5 inch offsets on either side of the wall.



brick or masonry foundation or strip footing

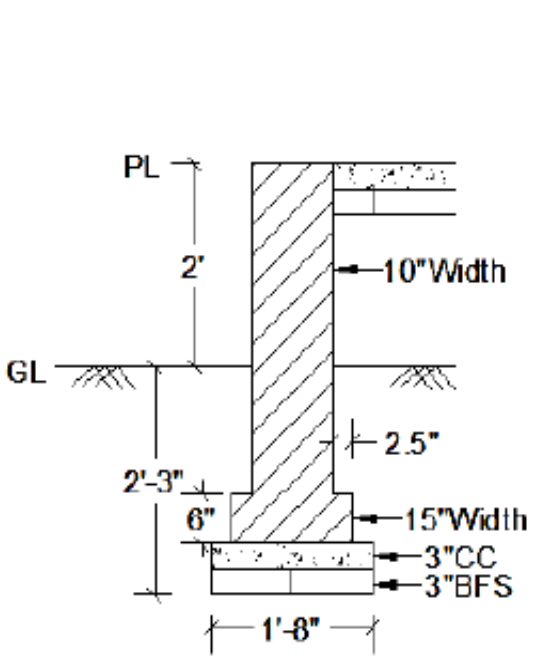


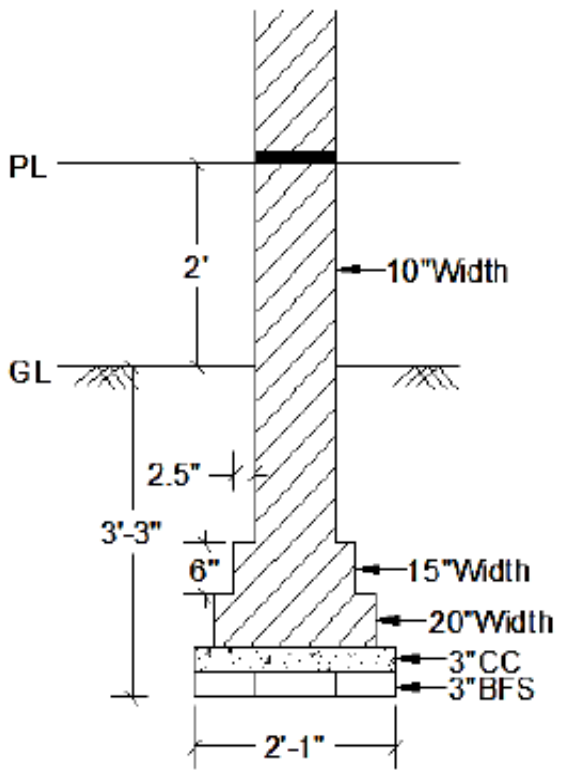
Estimate the materials required for the following residential building (elementary information is given)

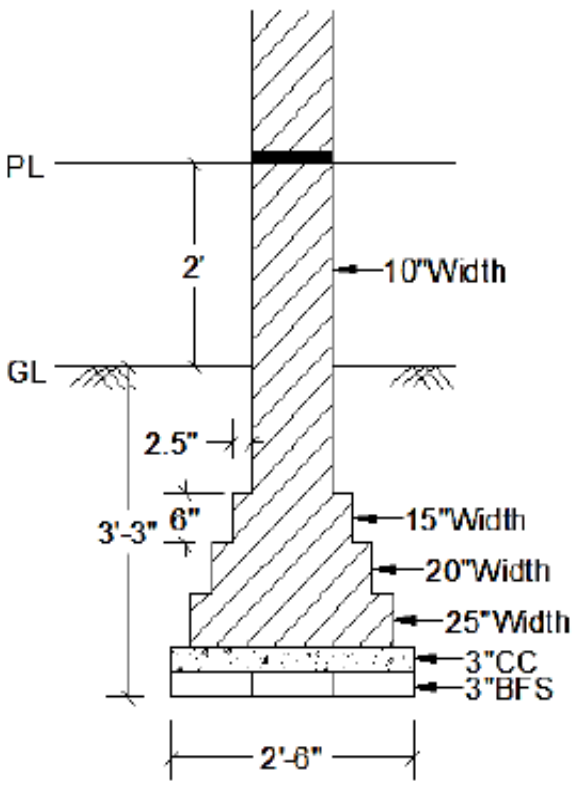


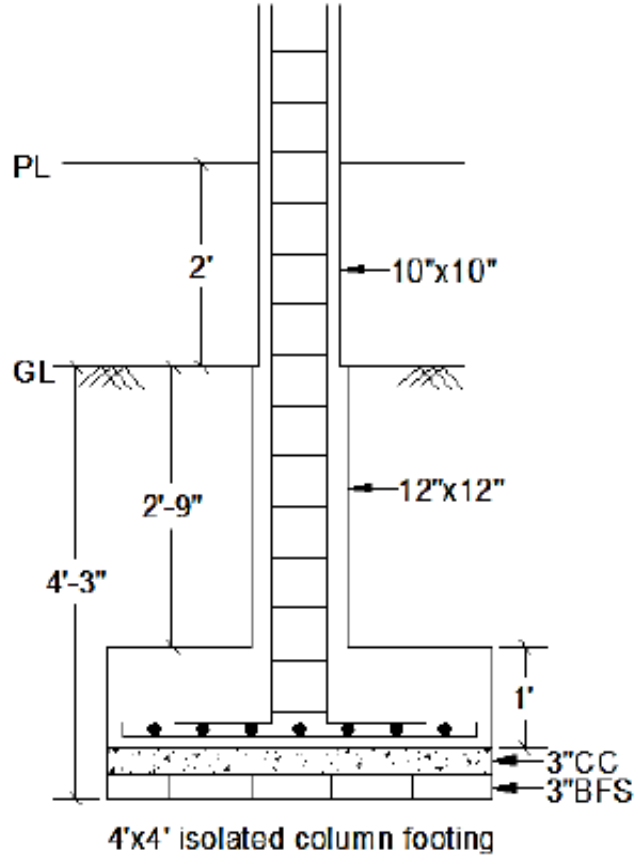
The four types of foundation used and their cross-sections

1. 30″ wall foundation
2. 25″ wall foundation
3. 20″ wall foundation
4. Column foundation (Isolated Column Footing)

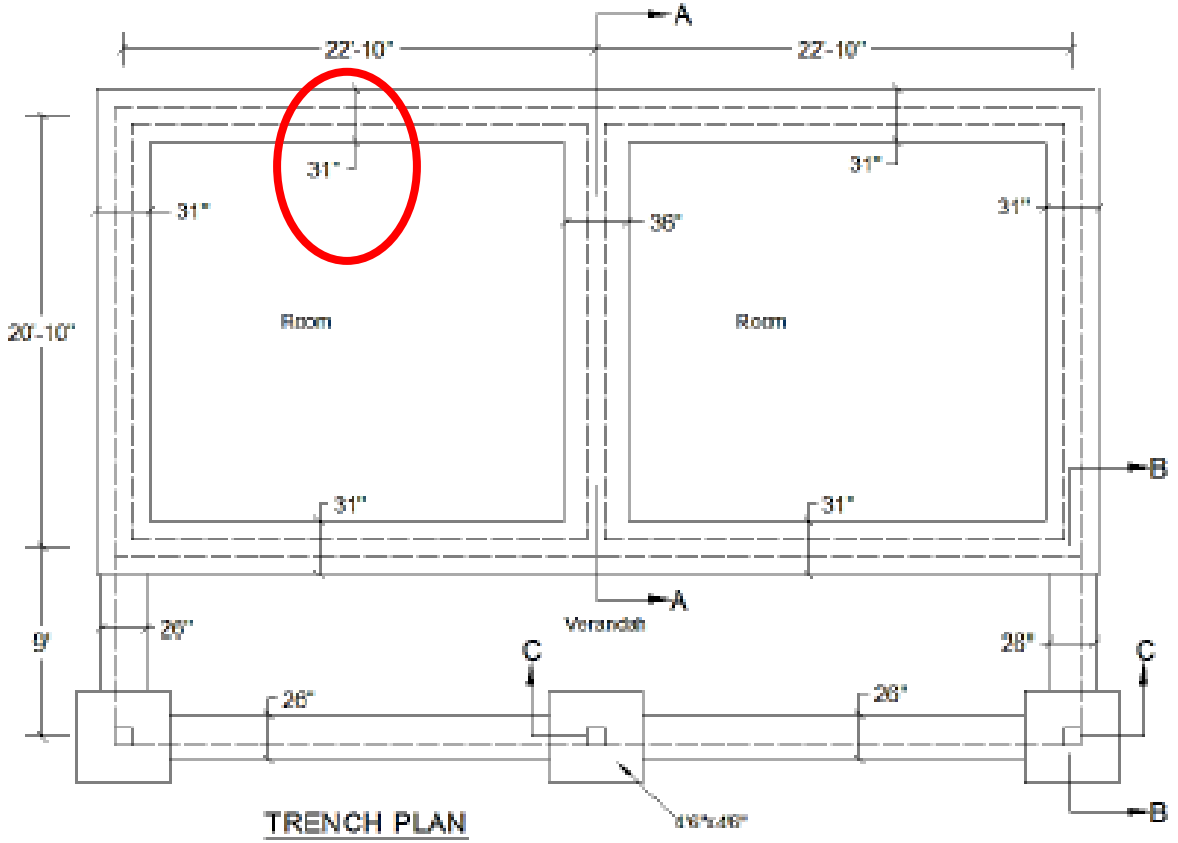


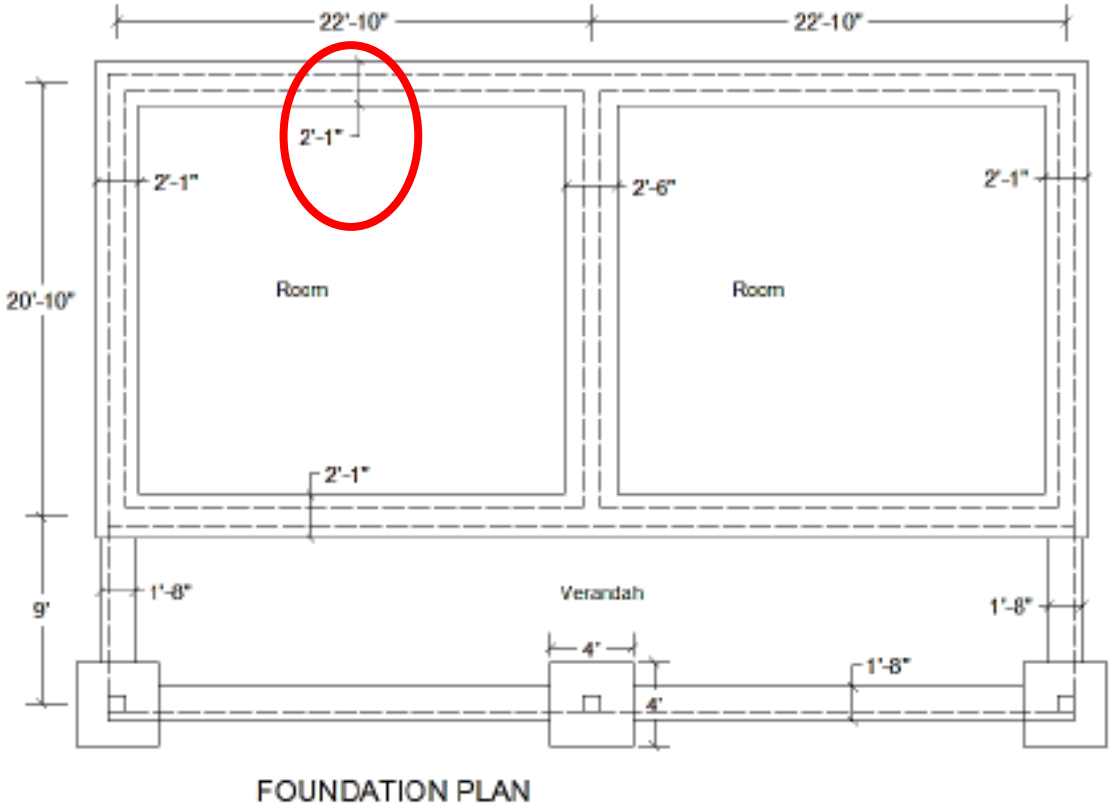






4'x4' isolated column footing



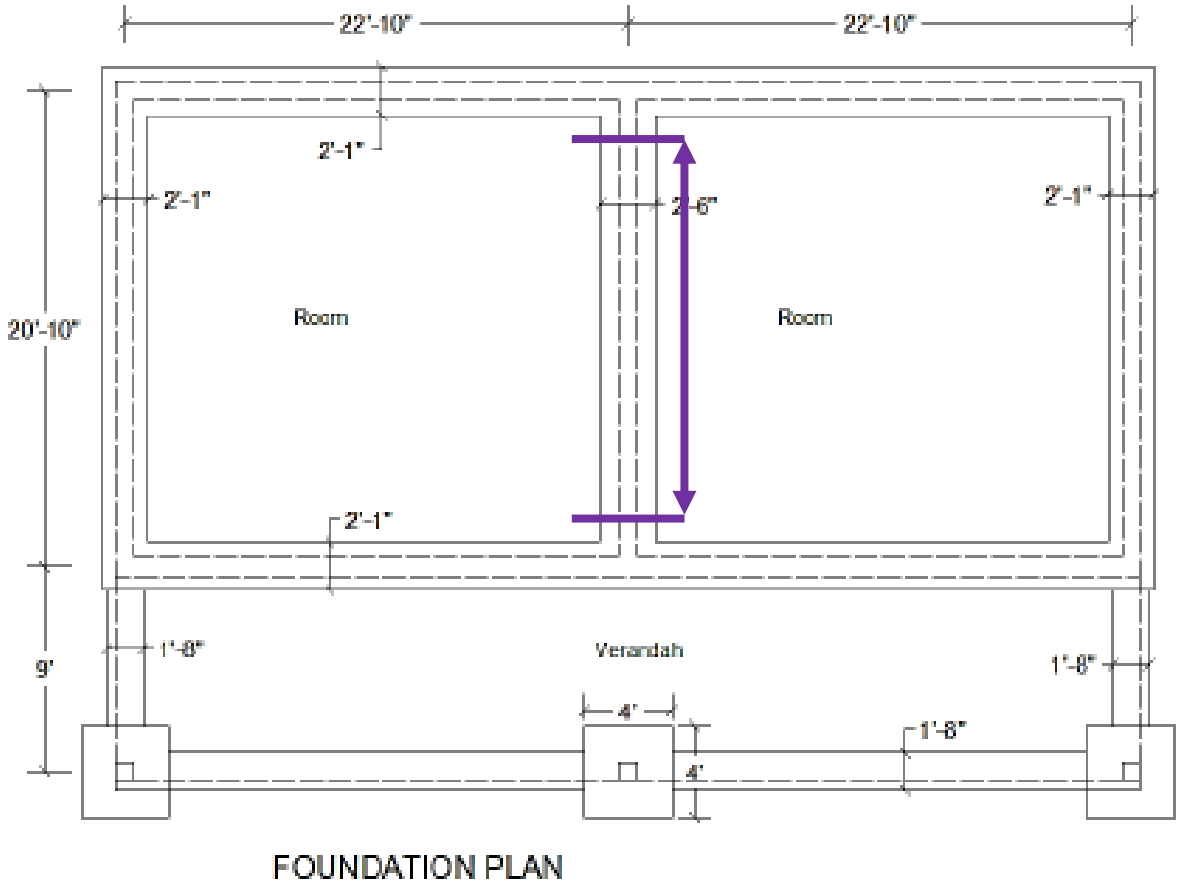




Extra width for excavation

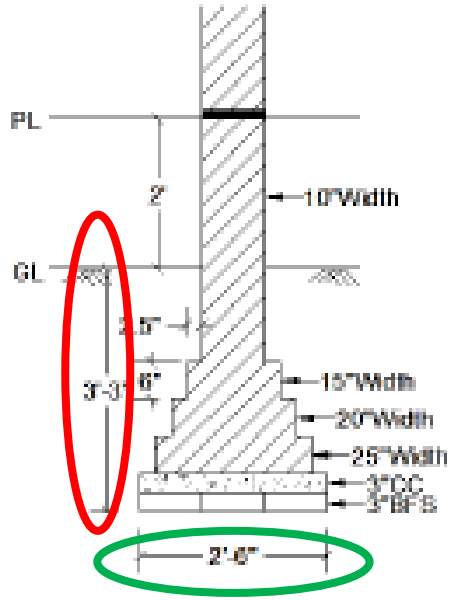
For excavation dimensions, the actual width of foundation is offsetted by **3 inch** on **all sides** (compare the foundation plan and trench plan). If trench plan is given, you can obtain the length of excavation directly from trench plan.

1. Estimation of Earthwork Excavation



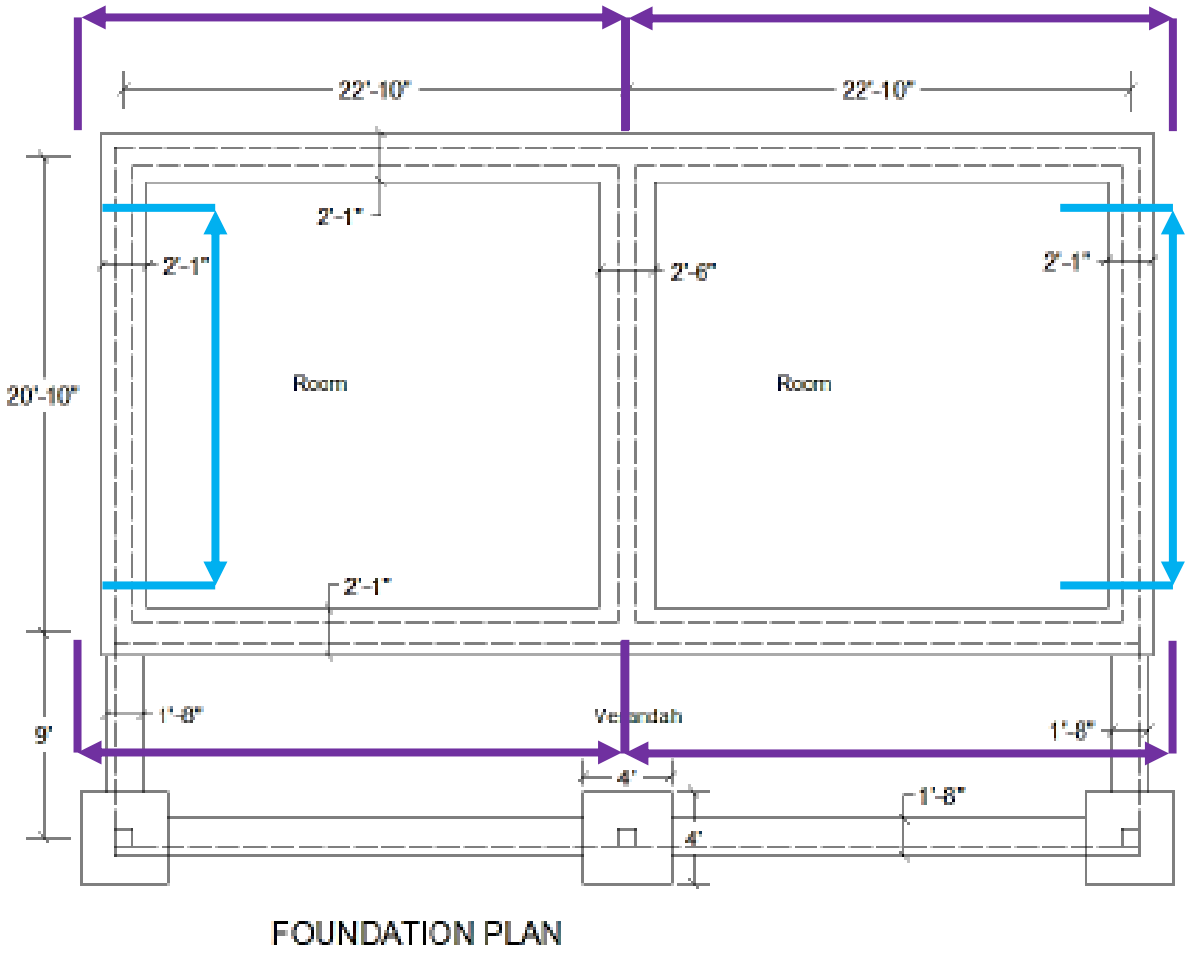
FOUNDATION PLAN

30" wall foundation:



- Height= 3'3"
- Width= (2'6")+3"+3"
- Length= 20'10"-12.5"-3"-12.5"-3"

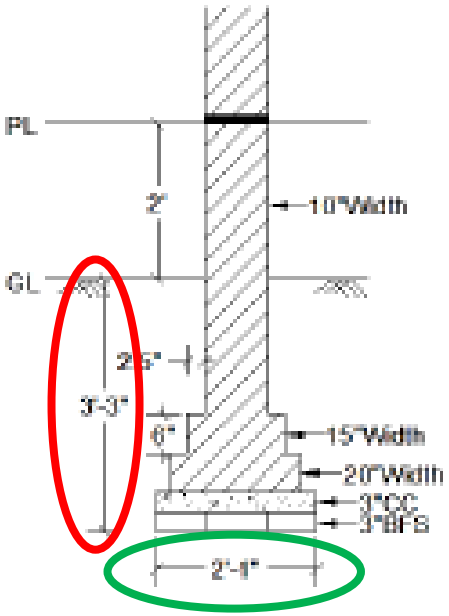
1. Estimation of Earthwork Excavation



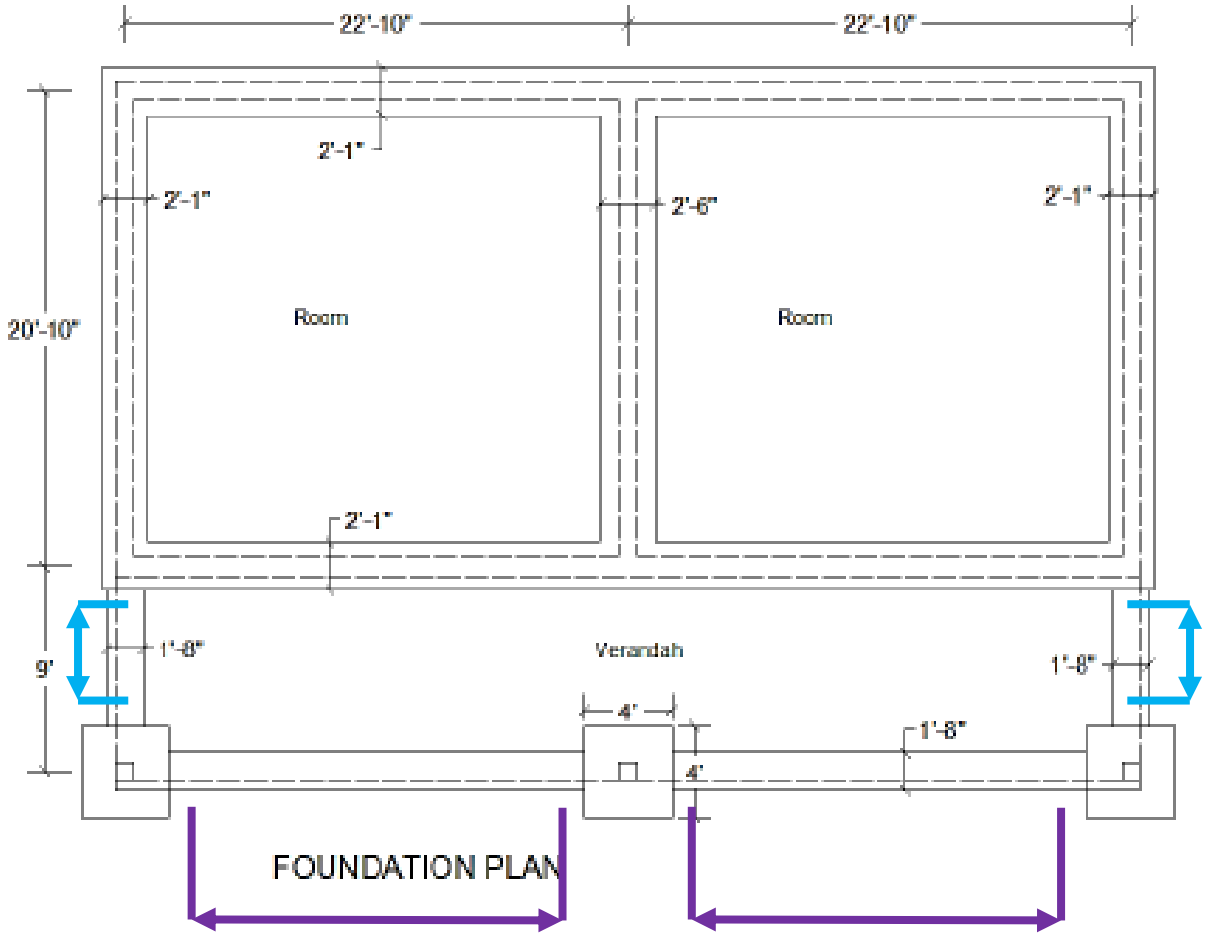
FOUNDATION PLAN

25" wall foundation:

- Height= 3'3"
- Width= (2'1")+3"+3"
- Length=((22'10"+12.5"+3")*4) +((20'10"-25"-6")*2)

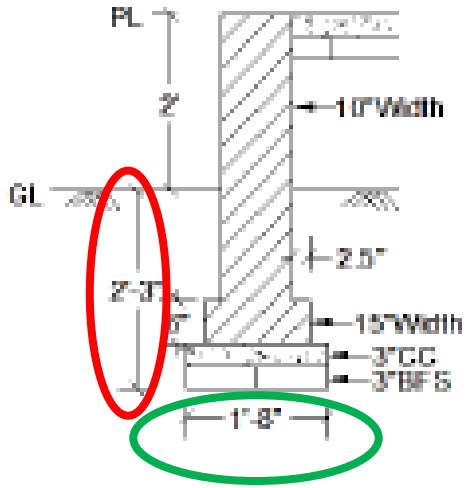


1. Estimation of Earthwork Excavation

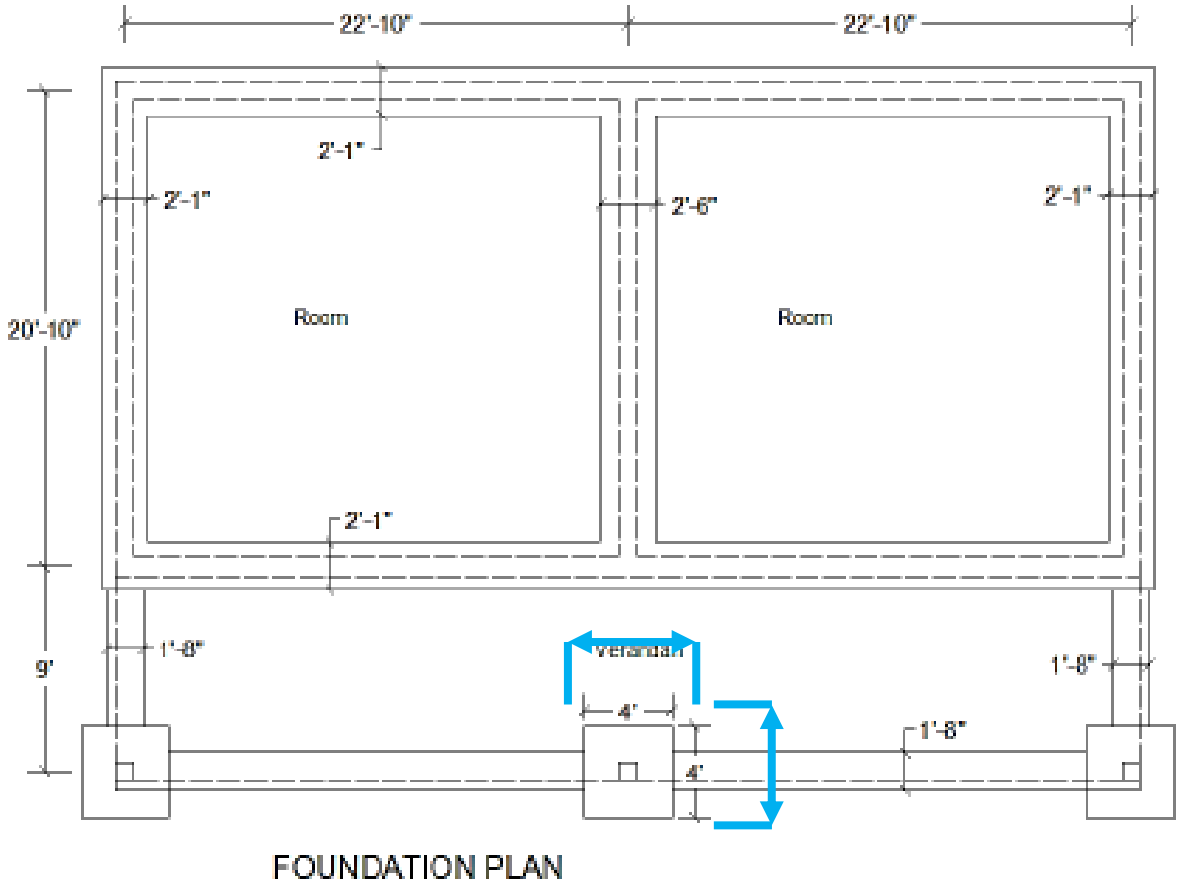


20" wall foundation:

- Height= 2'3"
- Width= (1'8")+3"+3"
- Length=((9'-2'-3"-12.5"-3")*2) + ((22'10"-4'-6")*2)

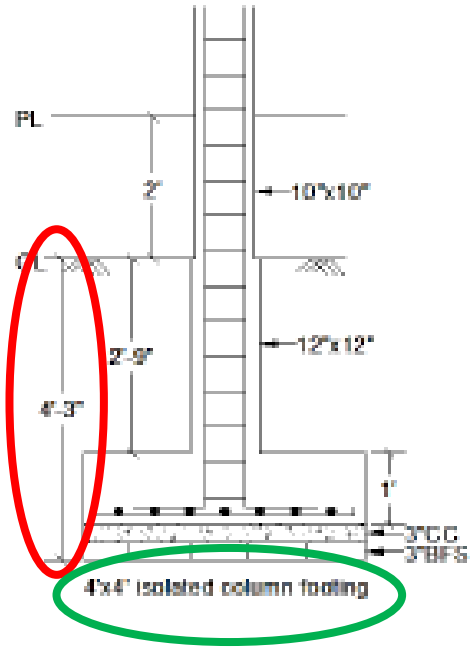


1. Estimation of Earthwork Excavation

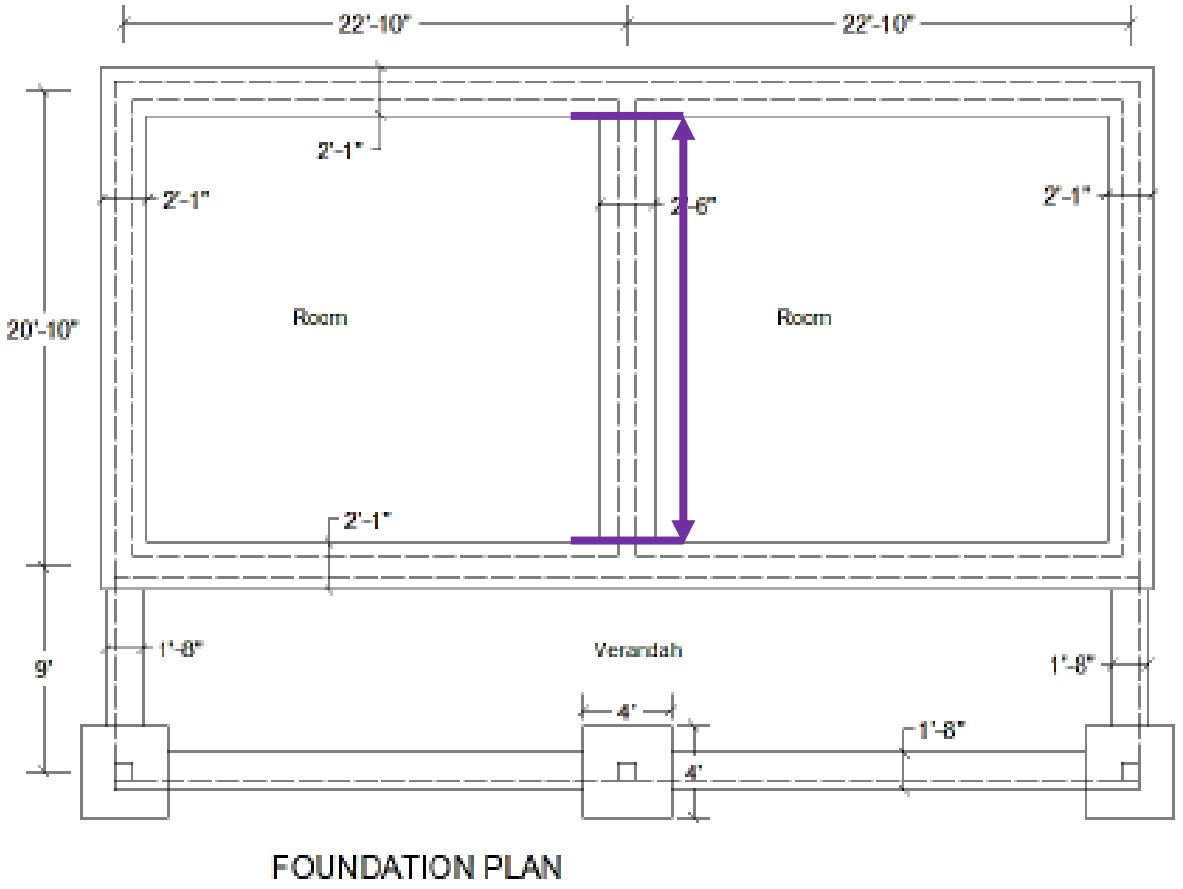


Column foundation:

- Height= 4’3”
- Width= 4’+3”+3”
- Length= 4’+3”+3”



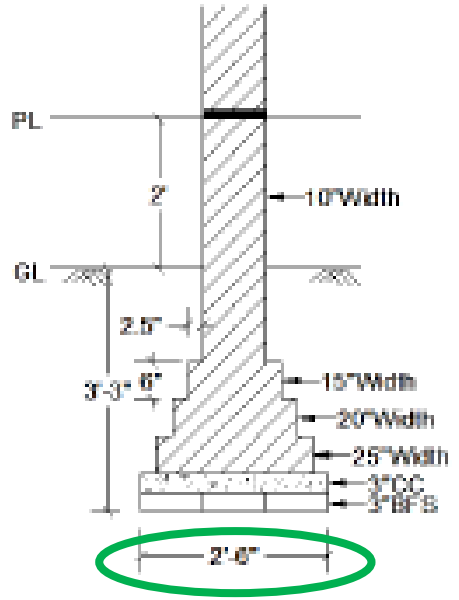
2. Estimation of Brick Flat Soling (one layer – 3“ thick)



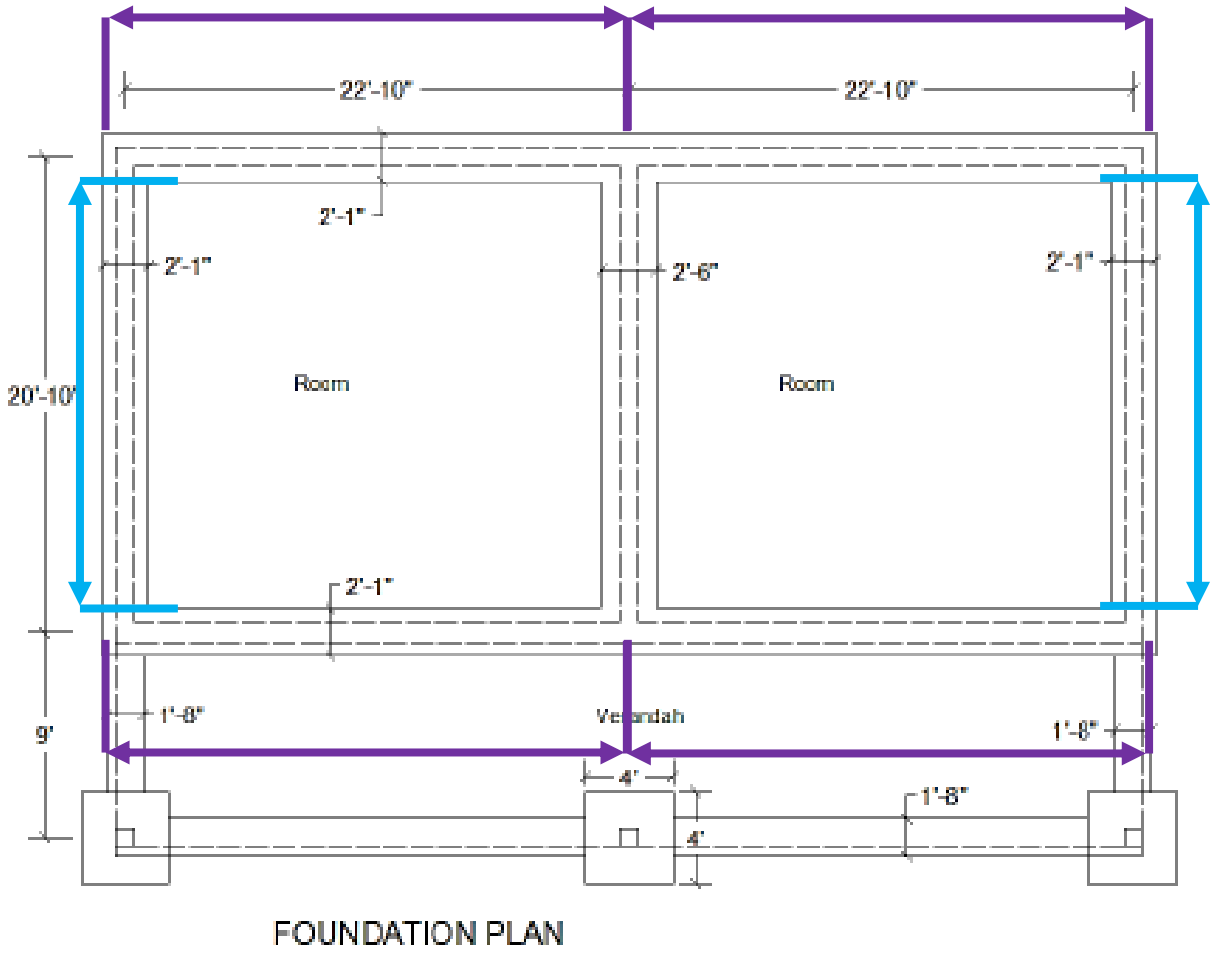
FOUNDATION PLAN

30” wall foundation:

- Width= 30”
- Length= 20’10”-25”



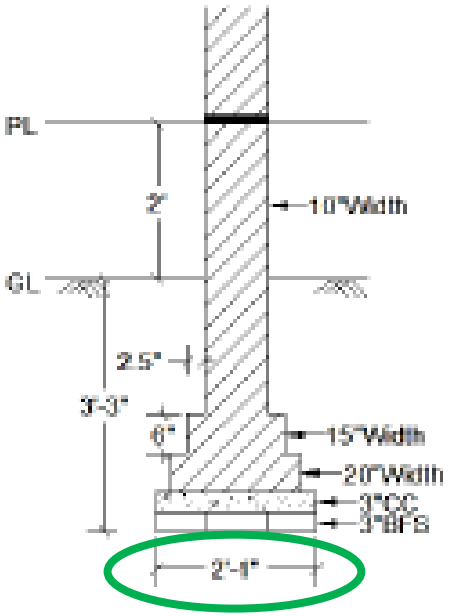
2. Estimation of Brick Flat Soling (one layer)



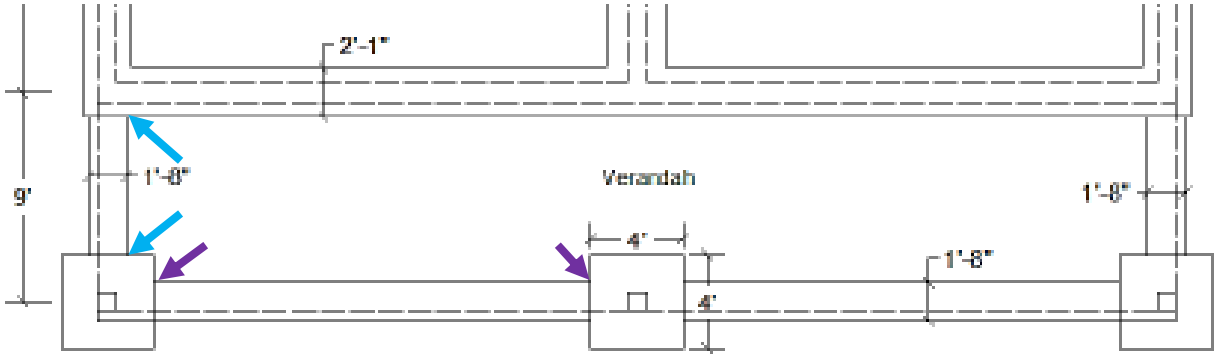
FOUNDATION PLAN

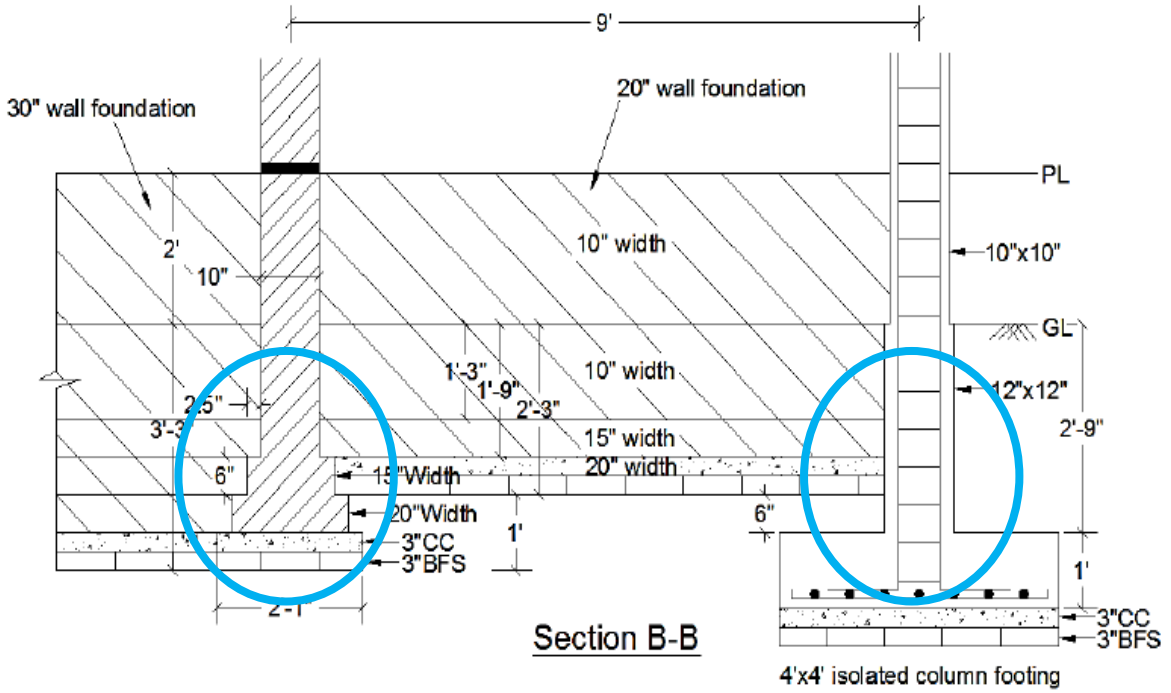
25" wall foundation:

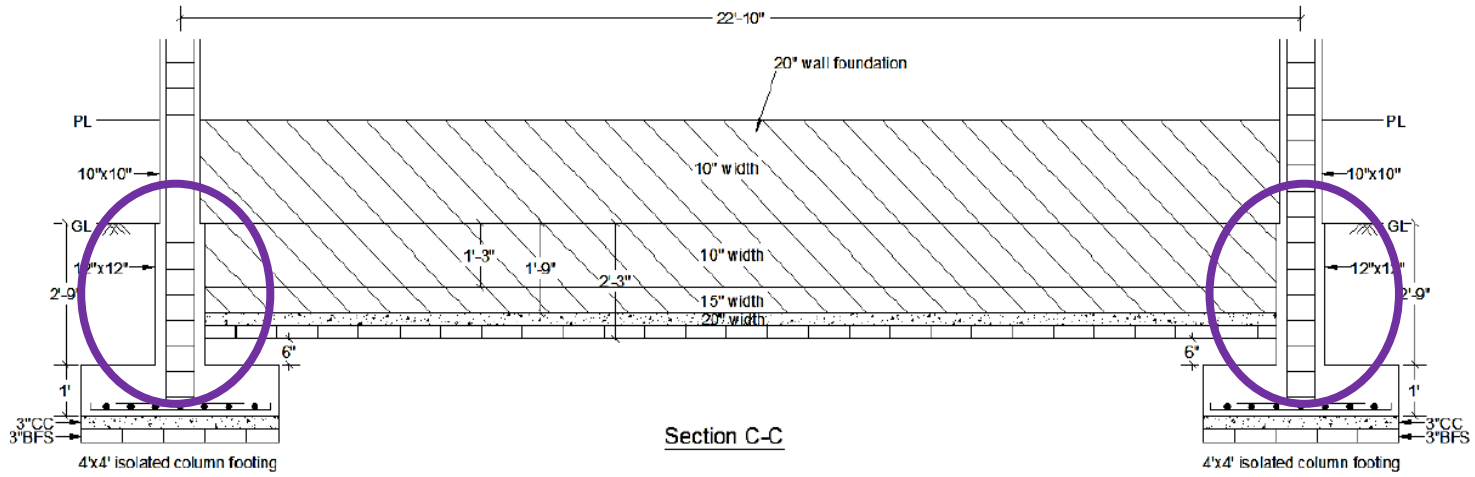
- Width= 25"
- Length=((22'10"+12.5") *4)+((20'10"-25")*2)



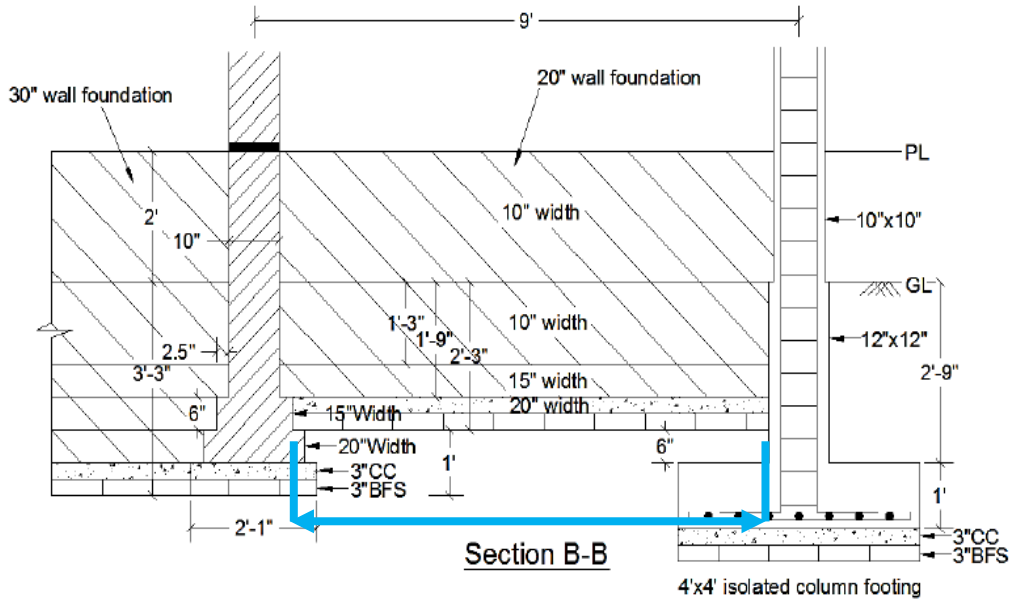
2. Estimation of Brick Flat Soling (one layer)

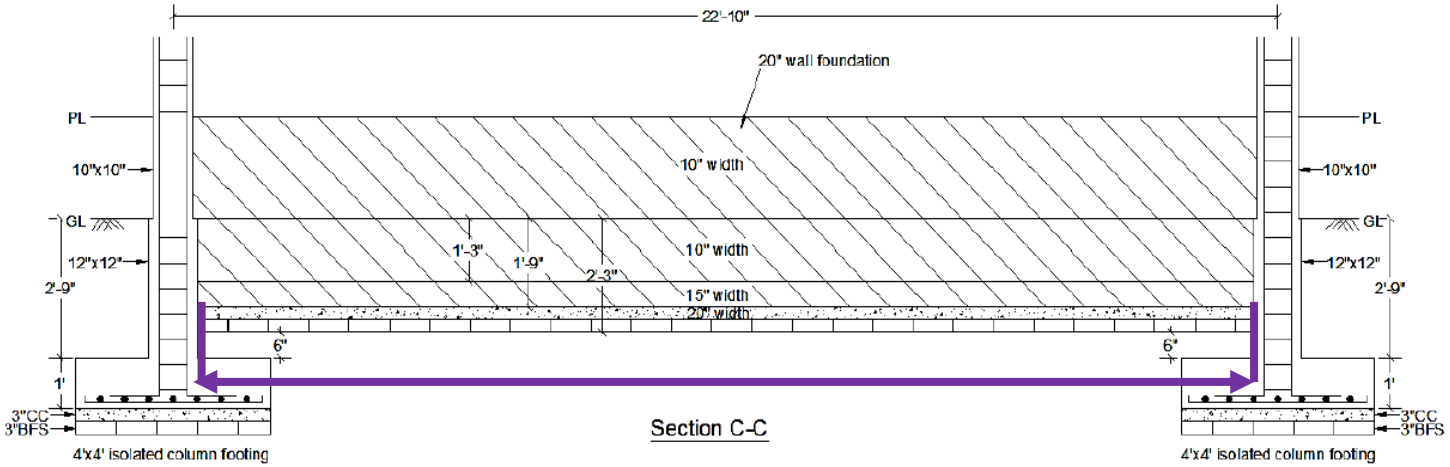






2. Estimation of Brick Flat Soling (one layer)

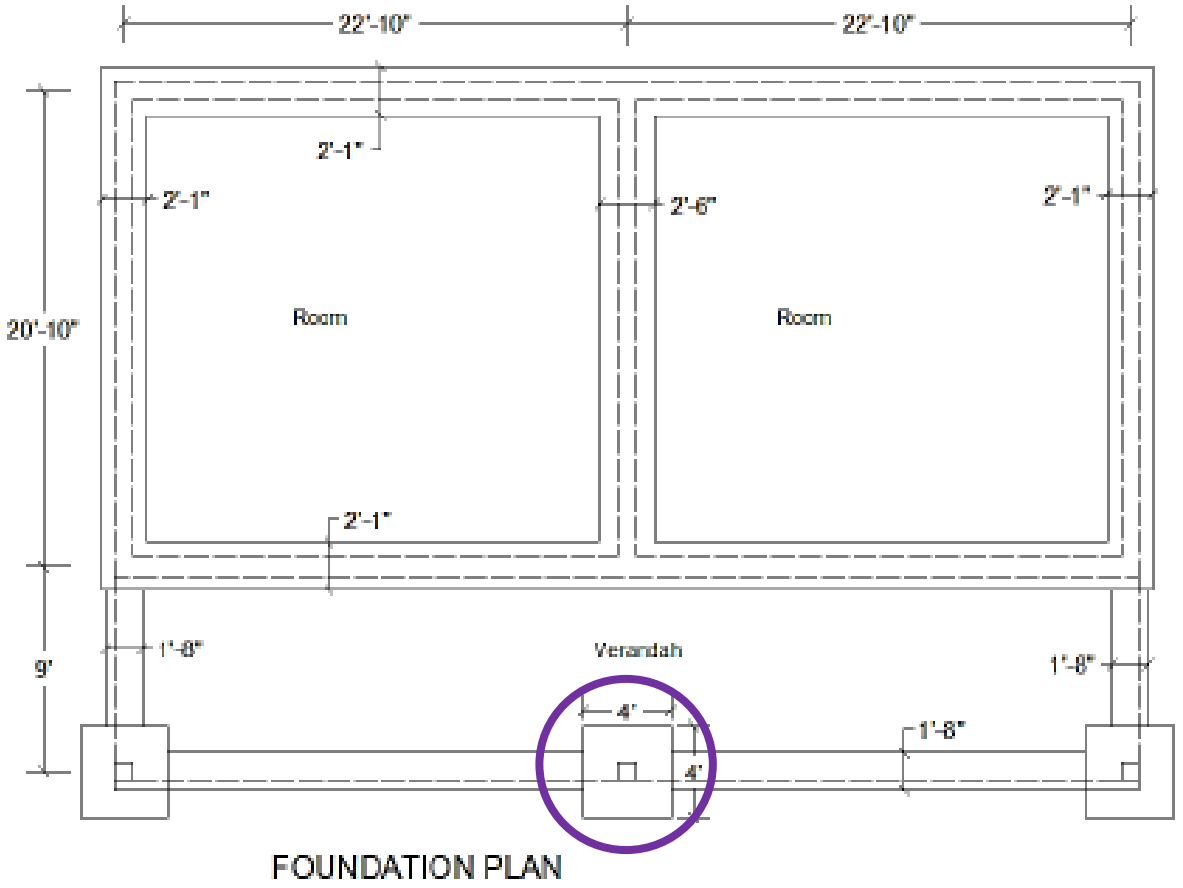




20” wall foundation:

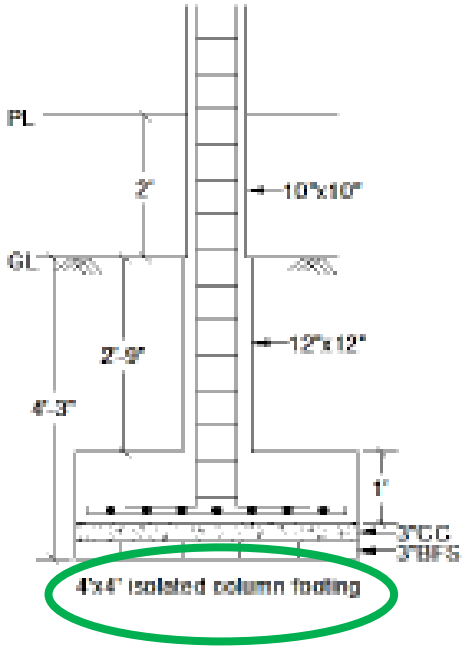
- Width= 20”
- Length=((9’-6”-7.5”)*2) + ((22’10”-12”)*2)

2. Estimation of Brick Flat Soling (one layer)

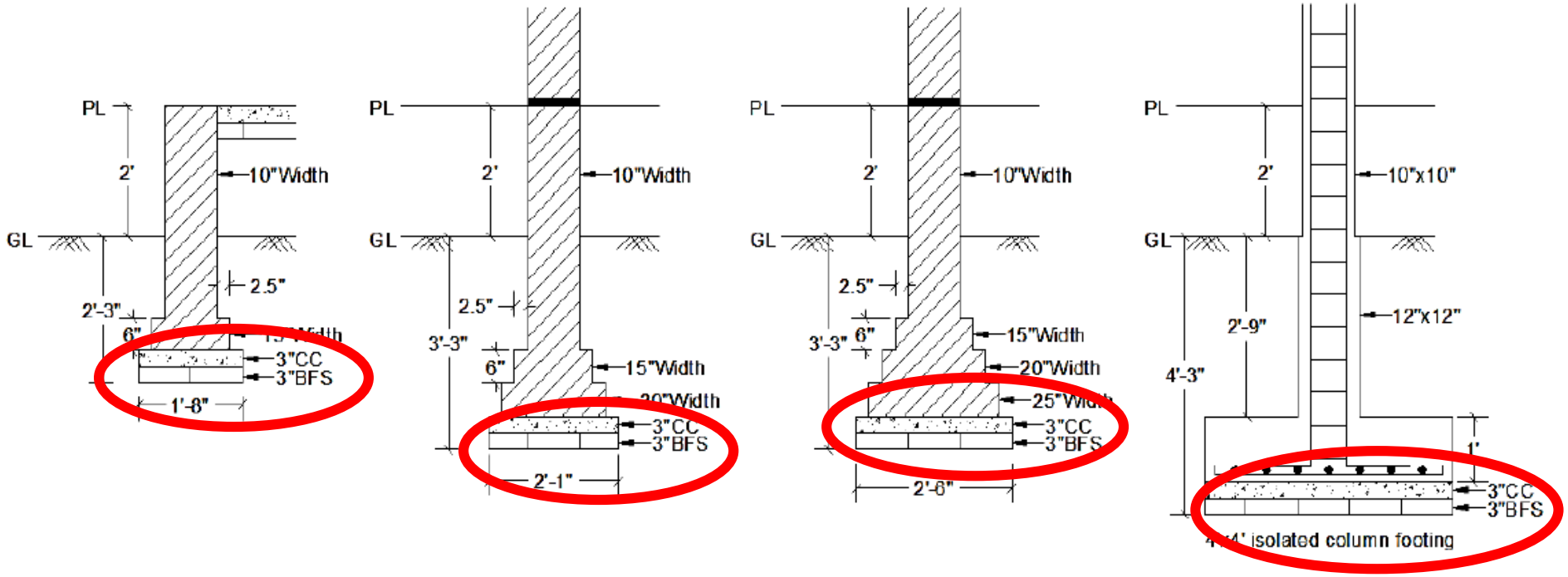


Column foundation:

- Width= 4’
- Length= 4’

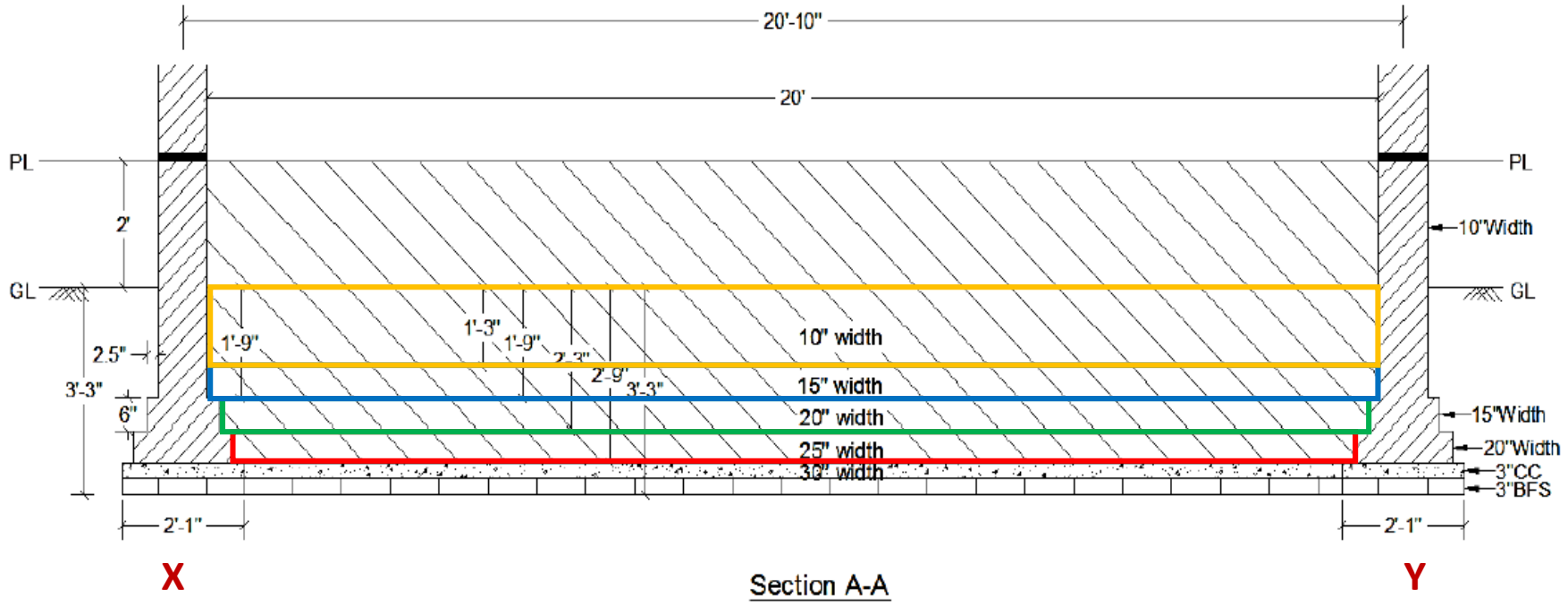


3. Estimation of Cement Concrete (1:3:6) in Foundation



- Length and width **same as BFS**!!!! For cement concrete, 3” thickness is to be considered for volume calculation

4. Estimation of Brickwork up to GL (30” wall foundation)



- The 30” wall foundation has a series of steps. At the bottom, its width is 30” and it gradually reduces to 10” decreasing by 5” in each step. The length along **XY** for each foundation width however increases by 5” as we move upwards

4. Estimation of Brickwork up to GL (30” wall foundation)

25” width:

- Height= 6”
- Width= 25”
- Length= 20’10”-25”+5”

20” width:

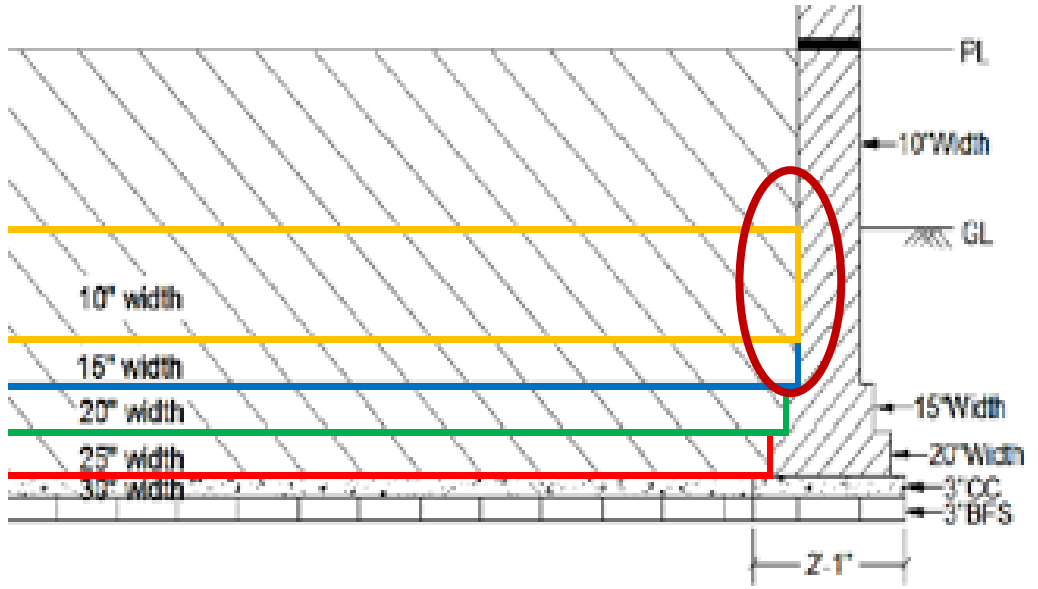
- Height= 6”
- Width= 20”
- Length= 20’10”-25”+5”+5”

15” width:

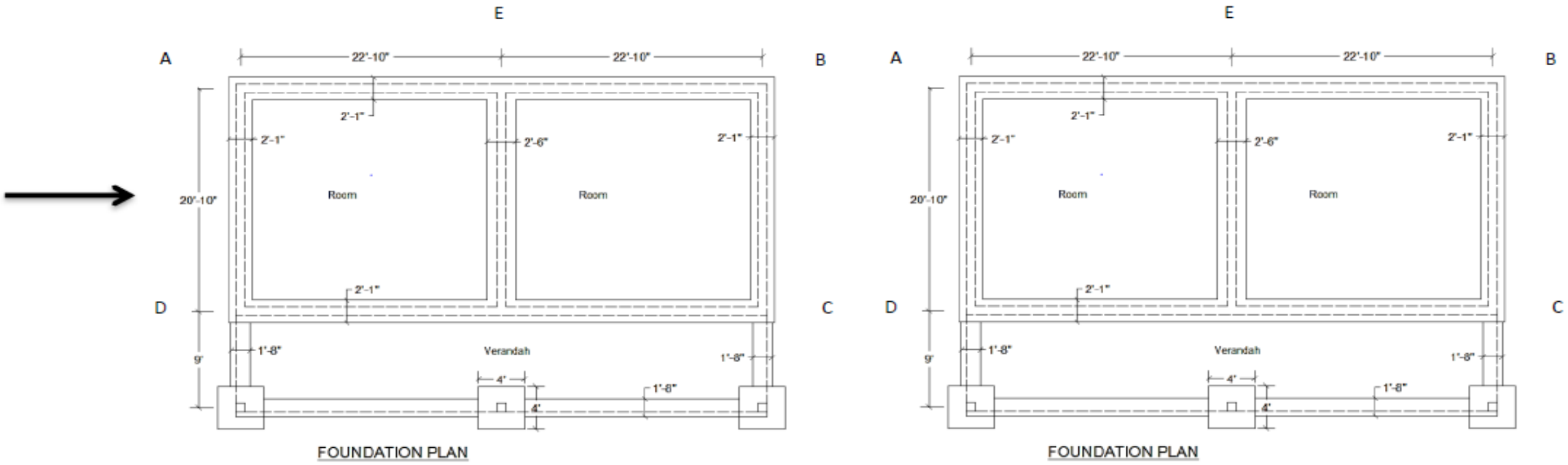
- Height= 6”
- Width= 15”
- Length= 20’10”-25”+5”+5”+5”

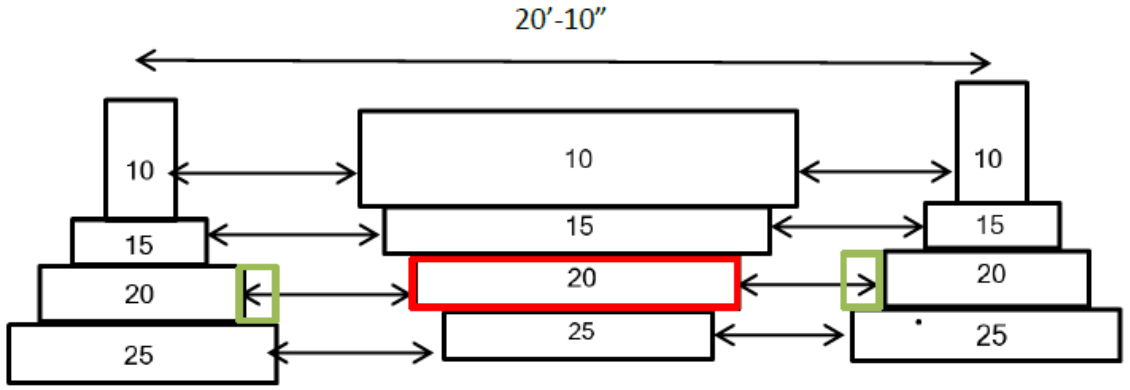
10” width:

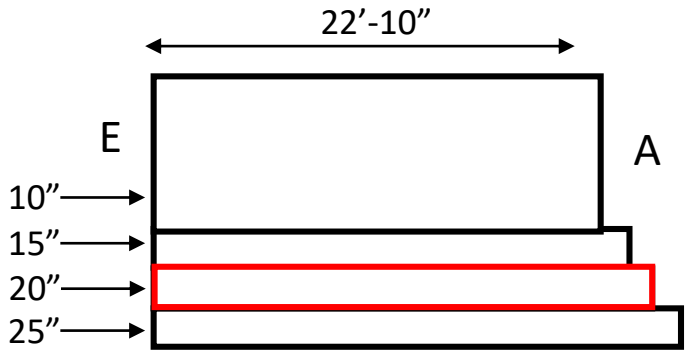
- Height= 15”
- Width= 10”
- Length= 20’10”-25”+5”+5”+5”



4. Estimation of Brickwork up to GL (25” wall foundation)







For 20” width, Length= ((22’10”+12.5”-2.5”)*4) + ((20’10”-25”+5”)*2)

4. Estimation of Brickwork up to GL (25” wall foundation)

Along AE, there is a **2.5”** decrease in length for 20” wall. However, along AD, there is a **5”** increase.

20” width:

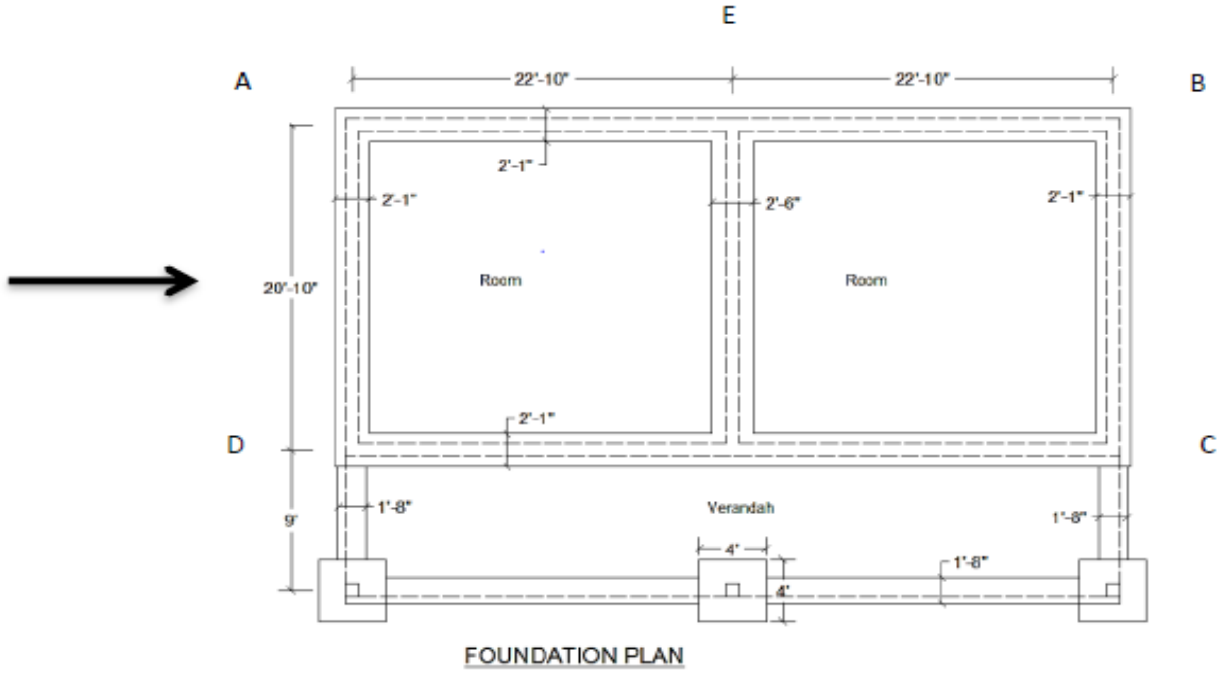
- Height= 6”
- Width= 20”
- Length= ((22’10”+12.5”-2.5”)*4) + ((20’10”-25”+5”)*2)

15” width:

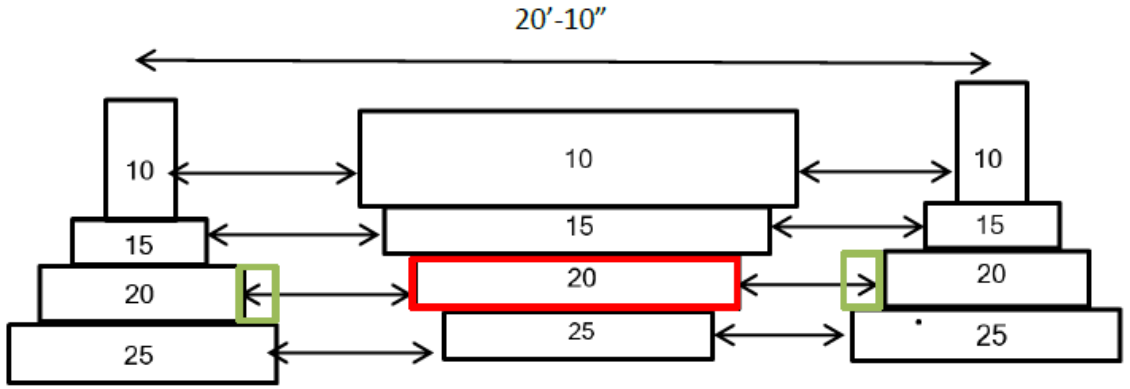
- Height= 6”
- Width= 15”
- Length= = ((22’10”+12.5”-5”)*4) + ((20’10”-25”+10”)*2)

10” width:

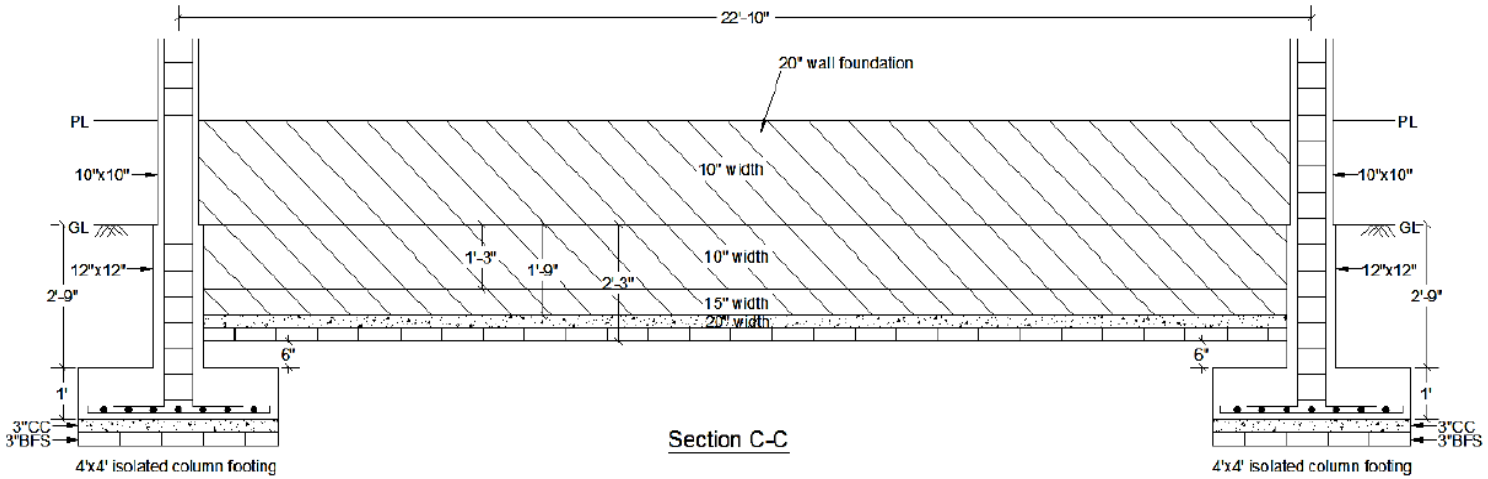
- Height= 21”
- Width= 10”
- Length= = ((22’10”+12.5”-7.5”)*4) + ((20’10”-25”+15”)*2)



FOUNDATION PLAN



4. Estimation of Brickwork up to GL (20” wall foundation)

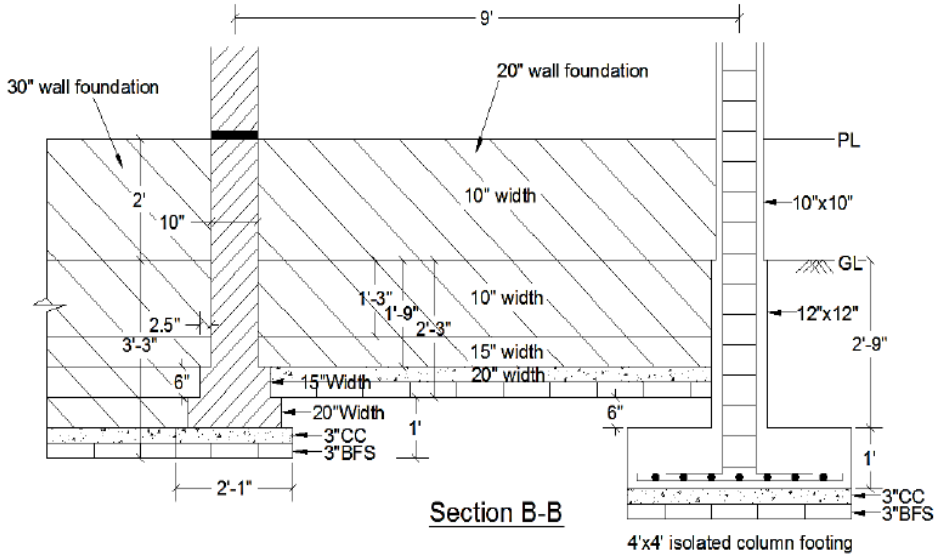


15” width:

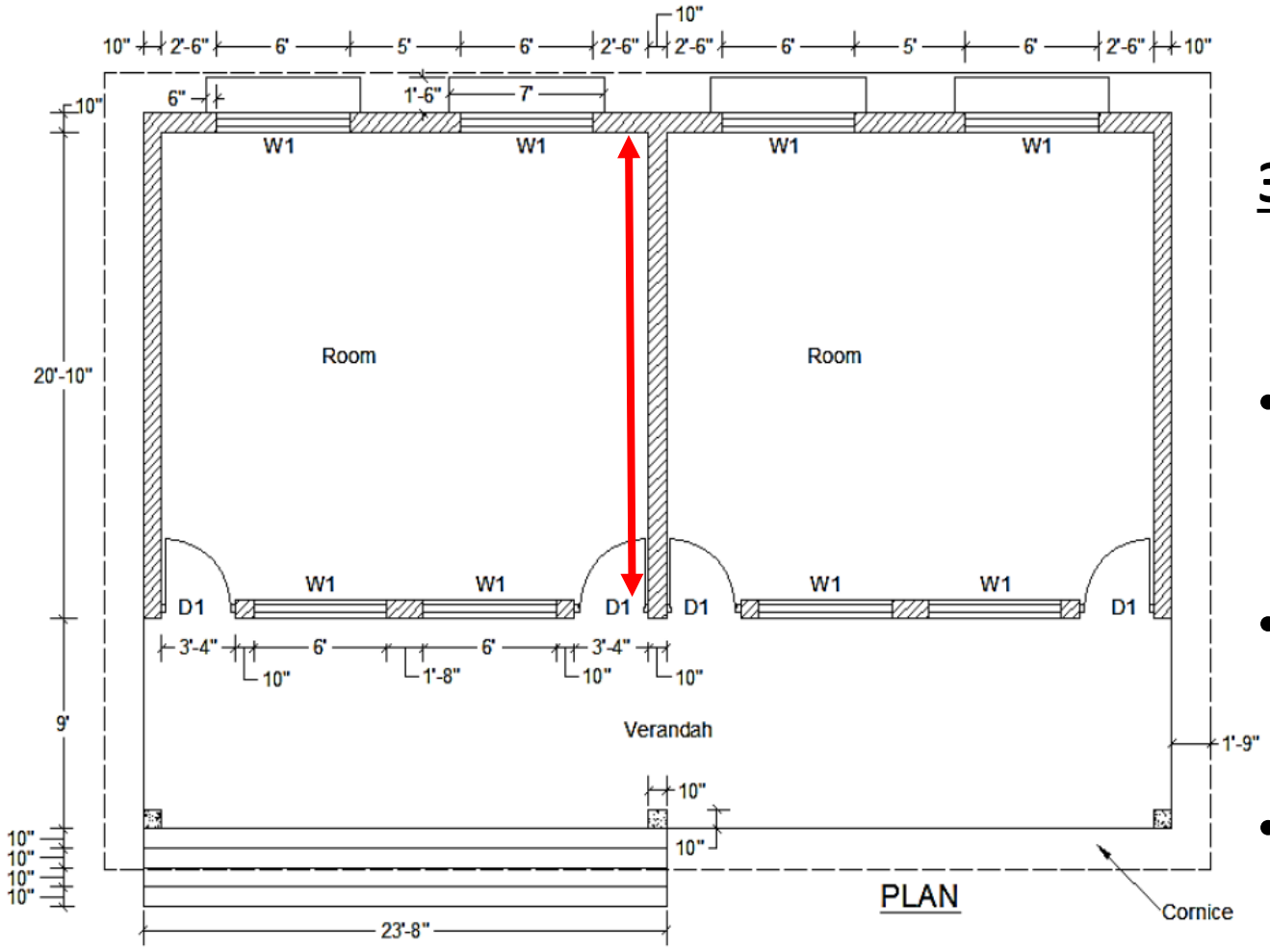
- Height= 6”
- Width= 15”
- Length= = ((9-6”-5”)*2) + ((22’10”-12”)*2)

10” width:

- Height= 15”
- Width= 10”
- Length= = ((9-6”-5”)*2) + ((22’10”-12”)*2)

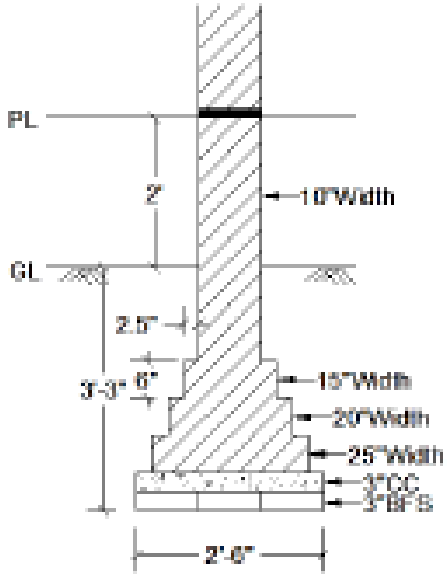


5. Estimation of Brickwork from GL to PL

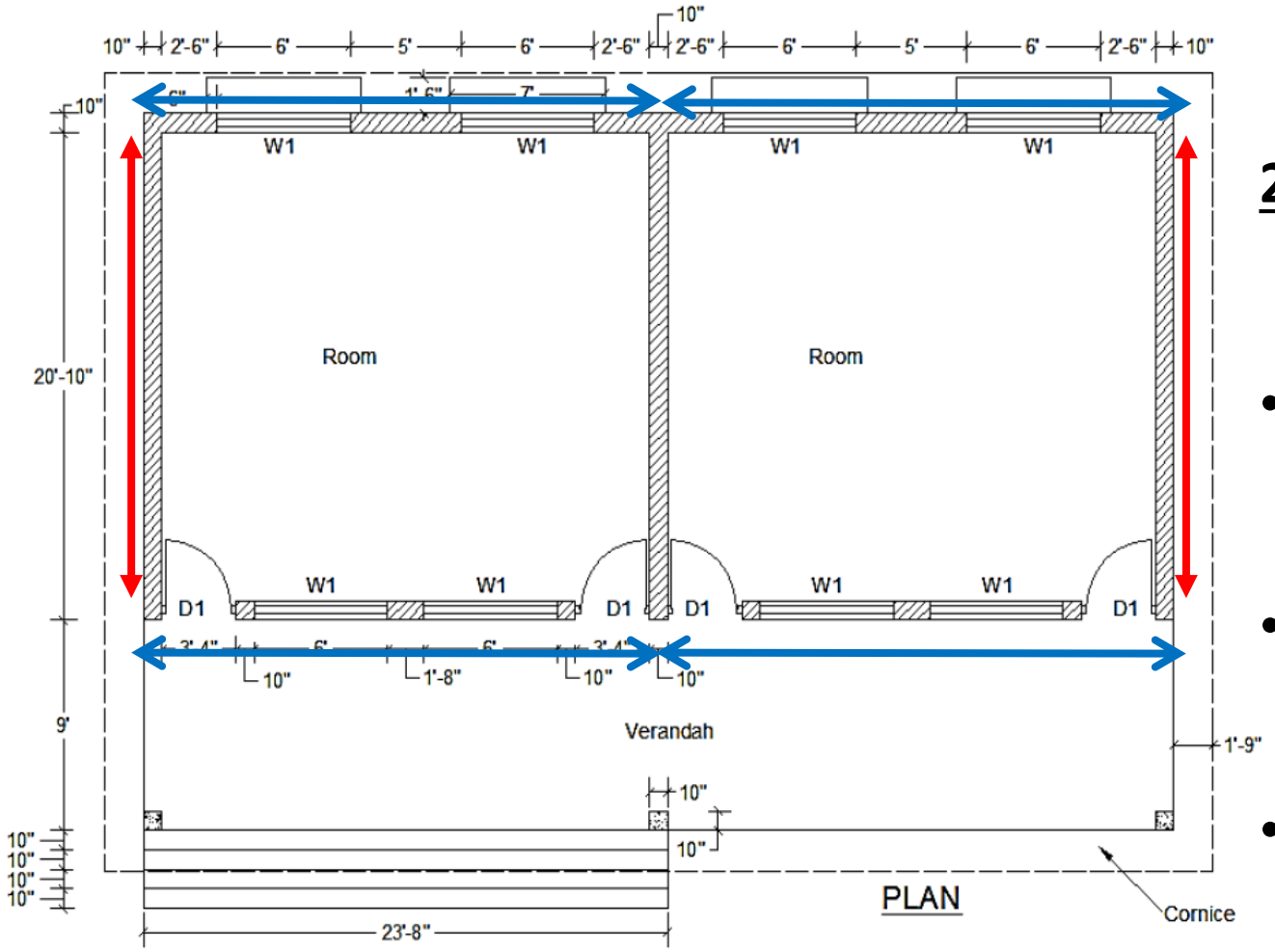


30” wall foundation:

- Height= 2’
- Width= 10”
- Length= 20’10”-10”

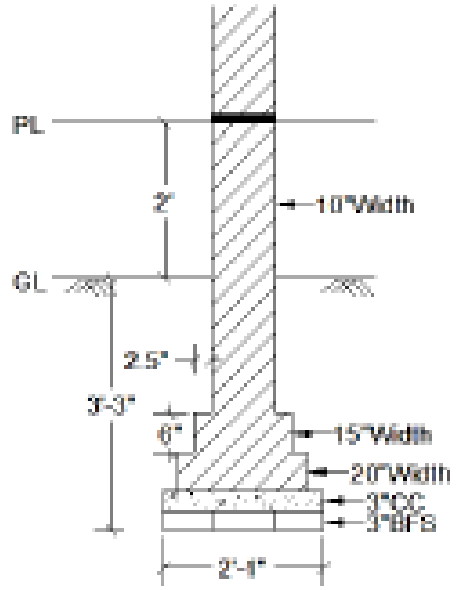


5. Estimation of Brickwork from GL to PL

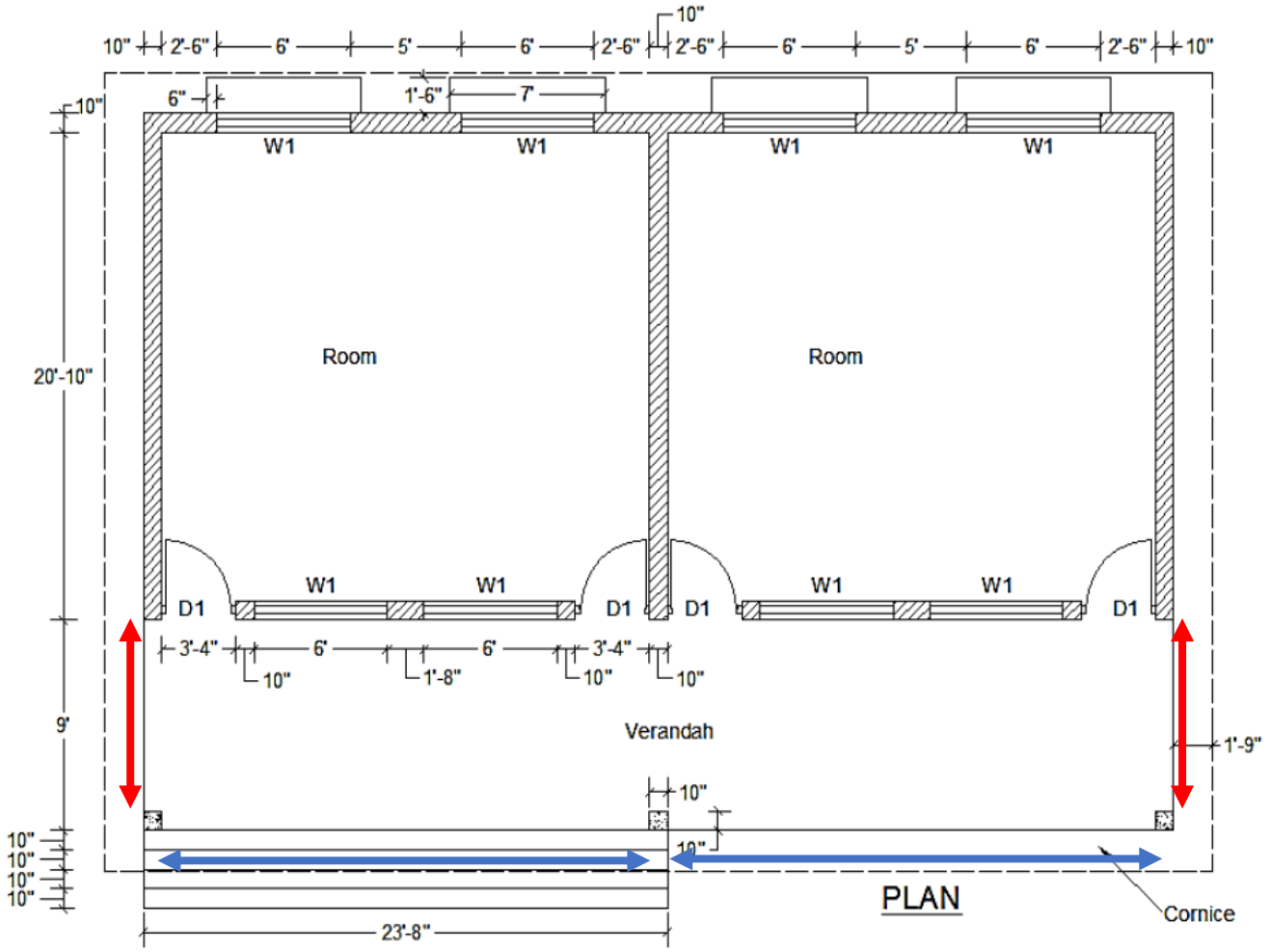


25” wall foundation:

- Height= 2’
- Width= 10”
- Length= (20’10”-10”)*2 +(22’10”+5”)*4



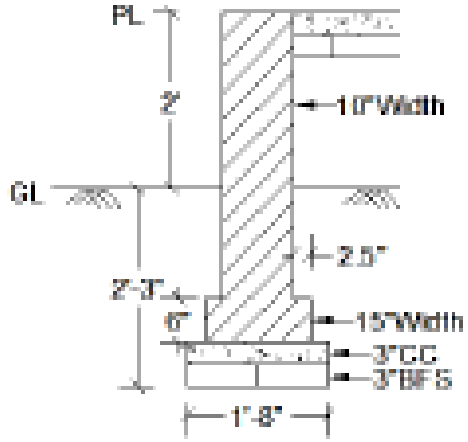
5. Estimation of Brickwork from GL to PL



20” wall foundation:

Height= 2’

Width= 10”



Length= (9’-10”)*2+(22’10”-10”)*2

6. RCC in Column (up to GL)

i) Base Slab

Volume= (4’x4’x1’)*3

i) Column

Volume= (1’x1’x1’9”)*3

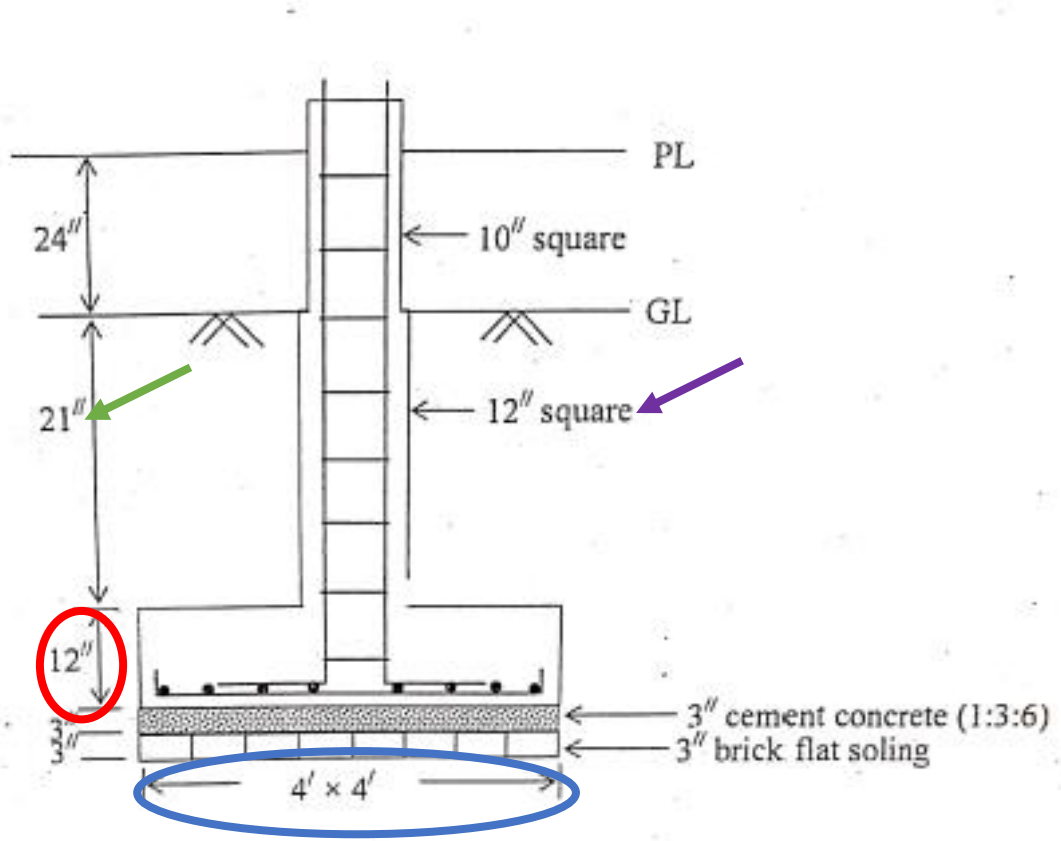


Figure: 4 ft x 4 ft isolated column footing

7. RCC in Column (GL to PL)

- Height= 2’
- Area= 10”x10”
- Volume= (2’x10”x10”)*3

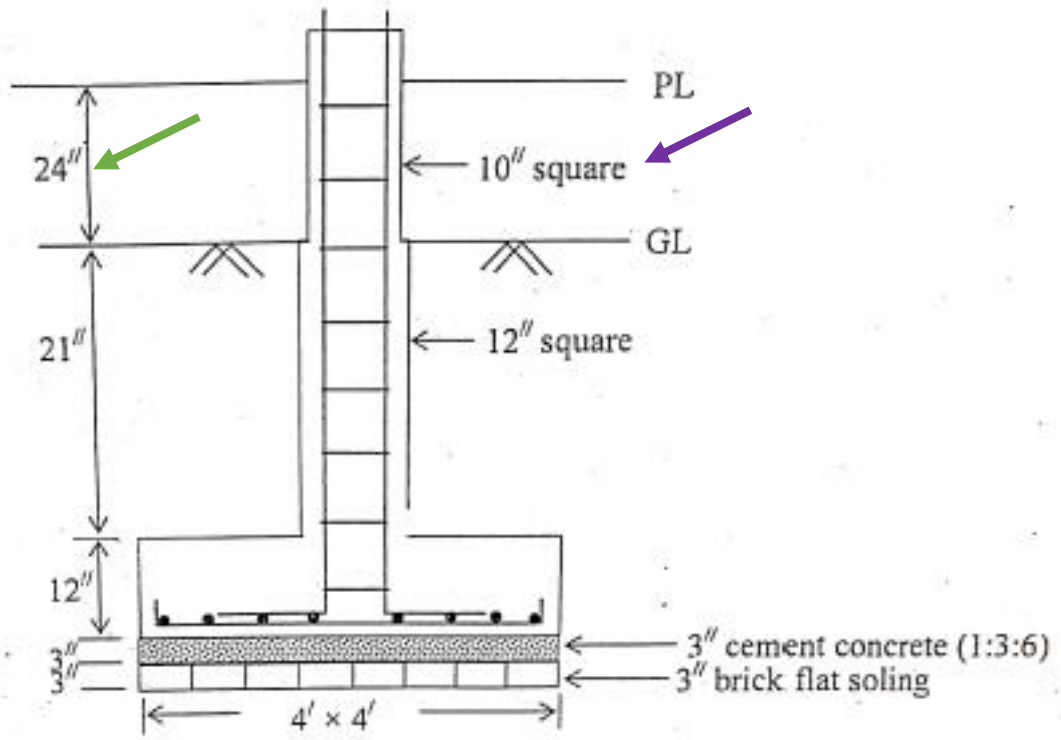


Figure: 4 ft x 4 ft isolated column footing

8. Damp Proof Course (DPC)

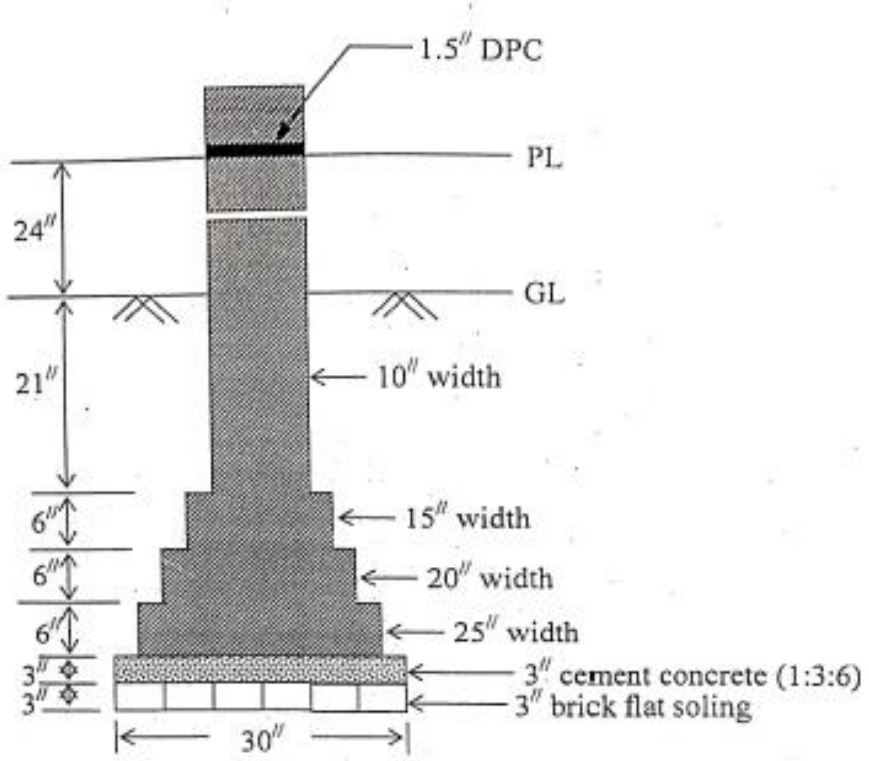


Figure: 30 inch continuous wall foundation

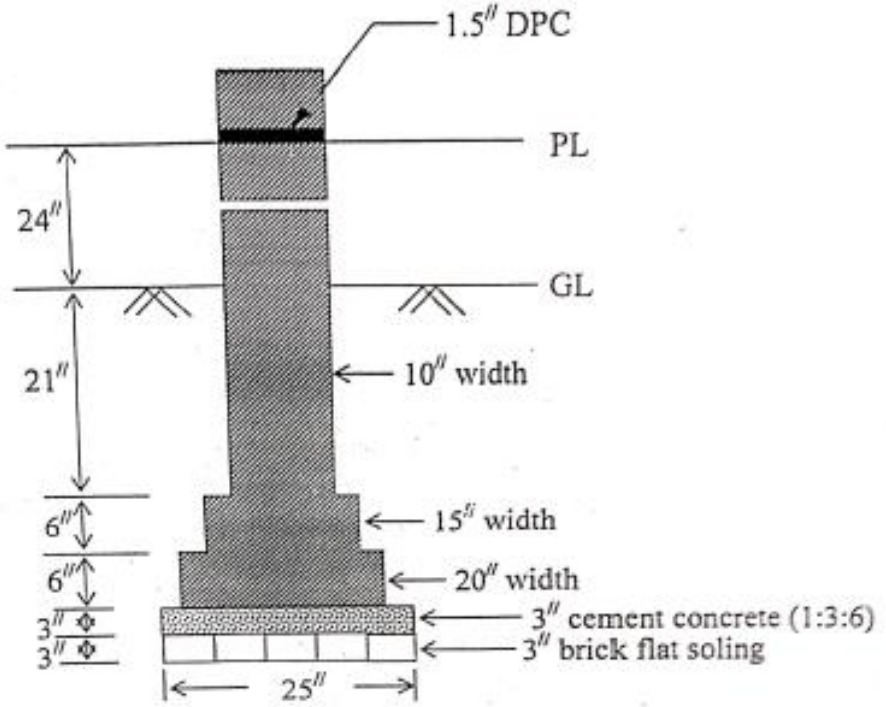


Figure: 25 inch continuous wall foundation

8. Damp Proof Course (DPC)

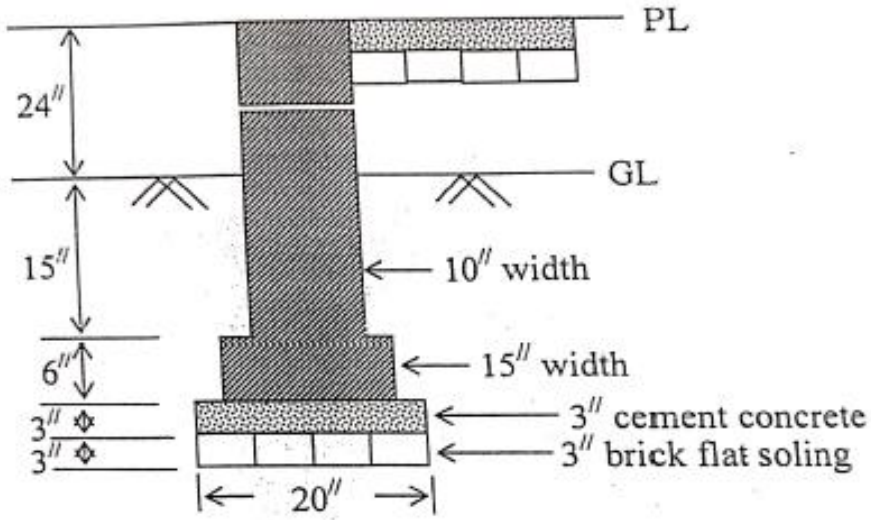


Figure: 20 inch continuous verandah wall foundation

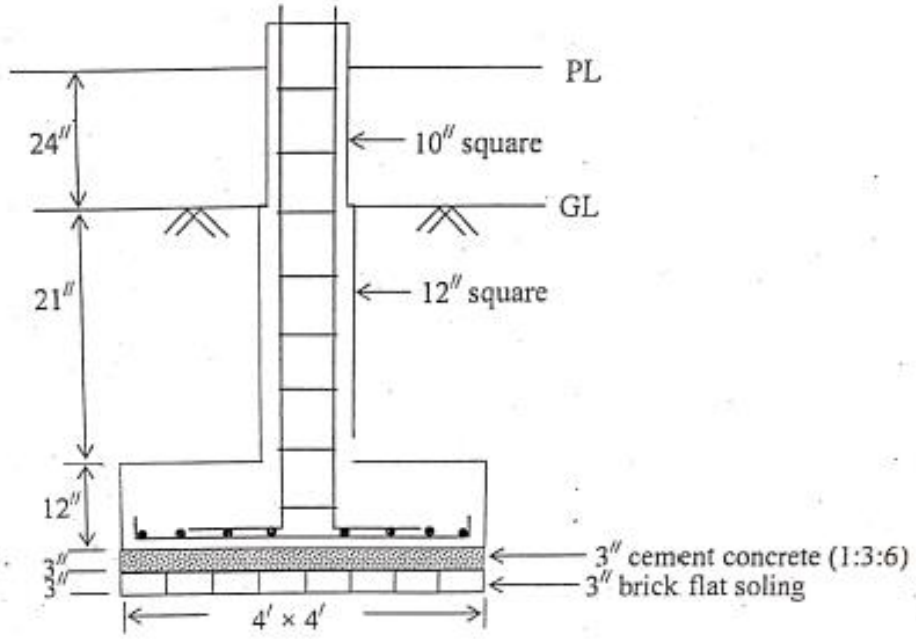
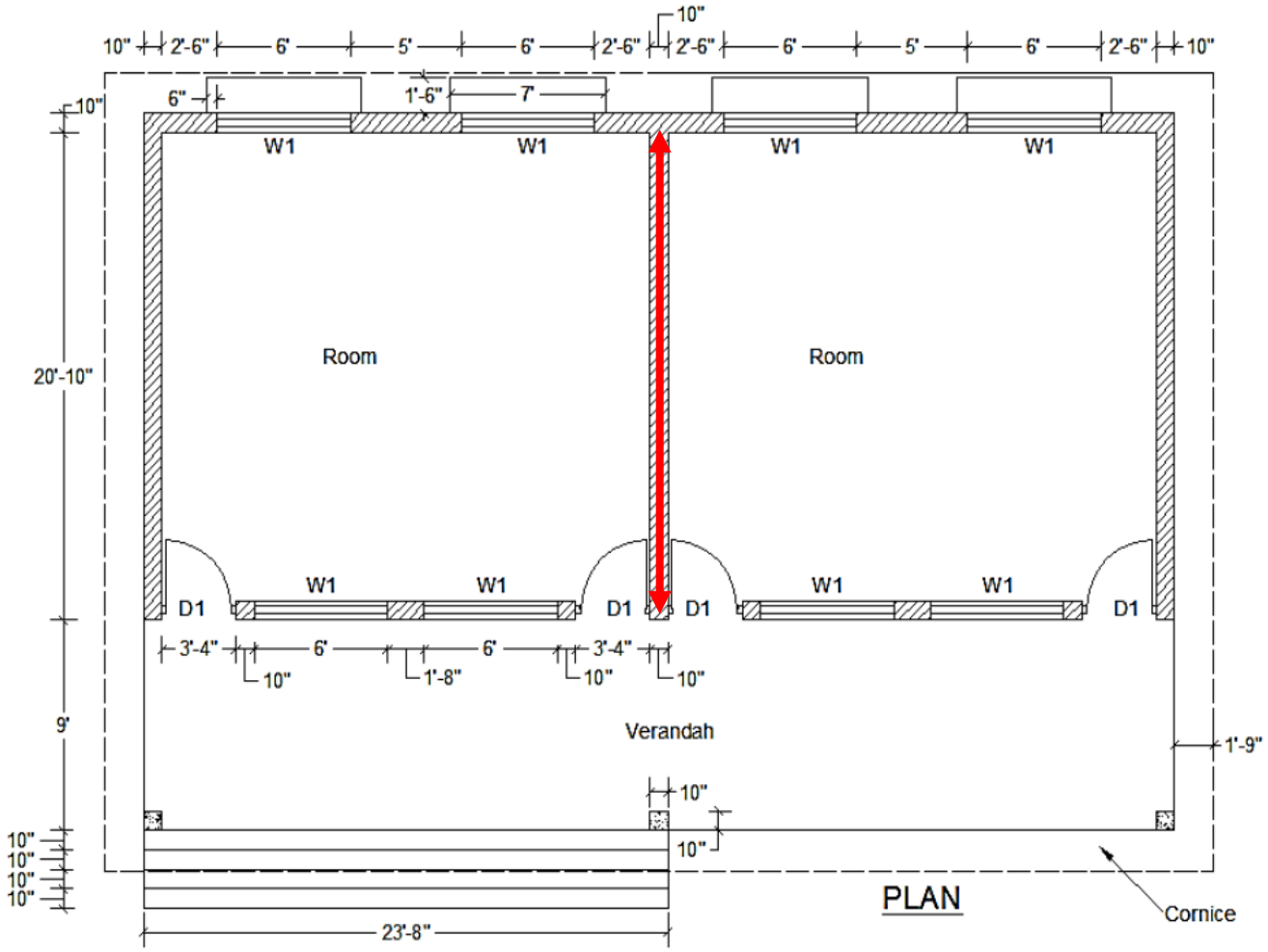


Figure: 4 ft x 4 ft isolated column footing

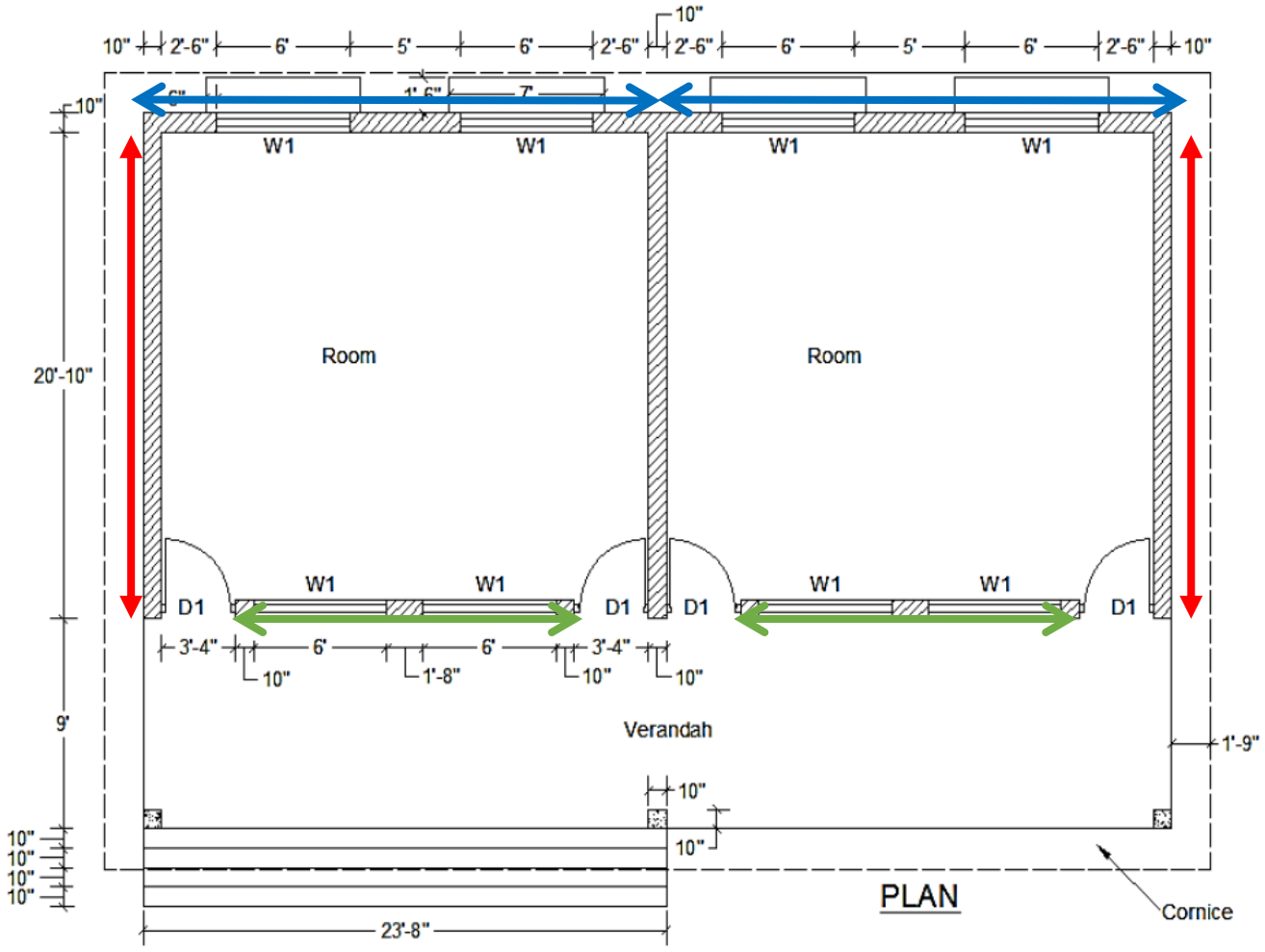
8. Damp Proof Course (DPC)



30” wall foundation:

- Thickness= 1.5”
- Width= 10”
- Length= 20’10”

8. Damp Proof Course (DPC)



25" wall foundation:

- Thickness= 1.5"
- Width= 10"
- Length= 20'10"*2+(22'10"+5")*2+(15'4")*2

Thank You!